Transform methods

1MA034

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Abstract

Lecture notes for a mathematical methods course given to various engineering track students. Main topics include:

- **Fourier series**. Periodic functions, complex and trigonometric Fourier series, even and odd functions, half-range series, basic L^2 theory and Parseval's theorem, pointwise convergence results.
- **Fourier transform**. Integral definition, scaling and shifting properties, convolution theorem, statement of Fourier inversion theorem and consequences, Plancherel's theorem.
- **Laplace transform**. Integral definition and motivation, transformation properties, initial and final value theorems, application to initial value problems, LTI systems.
- **Separable PDEs**. Separation of variables ansatz, the heat equation, the wave equation, inhomogeneous boundary conditions.
- **Z-transform**. Series definition, discrete convolution theorem, application to difference equations, discrete LTI systems.

Non-examinable material marked with ⋆.

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Recommended literature. These lecture notes overlap heavily with the notes by Samuel Edwards and Georgios Dimitroglou Rizell for other instances of this course. Some examples and derivations differ, but if you prefer to read Swedish notes I highly recommend them.

The only compulsory book for the course is:

- H. Sollervall and B. Styf, *Transformteori för ingenjörer*, Lund: Studentlitteratur (2006).

It covers (almost) all topics in the course at a level accessible for non-mathematicians. For those looking for a more mathematical, yet still accessible, discussion of the topics covered in chapters 1, 2 and 4, I recommend the following set of lecture notes.

- D. Skinner, *Mathematical Methods*, lecture notes (2015).

You may also be interested in chapters 19 and 20 of the (gargantuan) book:

- G.B. Arfken, H.J. Weber and F.E. Harris, *Mathematical methods for physicists (7th edition)*, Academic Press/Harcourt, Burlington, MA (2012).

While it may be somewhat overkill for this course, the book covers many topics you will undoubtedly see in other courses and will serve you well there. An online version is freely available through the Uppsala library.

Sections marked with a gold star ⋆ contain non-examinable content and can be safely skipped. They do, however, add some context to the material in that chapter and may be of interest to many students.

Caution! Notes may contain (many) typos. Feel free to email me if you find any.

Table of notation

symbol	definition
$\mathbb{R}, \mathbb{C}$	the field of real numbers and complex numbers respectively
$\mathbb{R}_{\geq 0}$	the set of real numbers satisfying $x \geq 0$
$\mathrm{Re}(z)$, $\mathrm{Im}(z)$	real and imaginary parts of $z \in \mathbb{C}$ respectively
$\bar{z}$	complex conjugate of $z \in \mathbb{C}$
$\log s$	the natural logarithm of $s \in \mathbb{R}$
$\chi_D(t)$	indicator function of the subset $D \subseteq \mathbb{R}$
$\sim$	states the Fourier series of a periodic function
c_n, $\{a_n, b_n\}$	the complex and trigonometric Fourier coefficients respectively
$S_N(t)$	the Nth partial sum of a Fourier series
Ω	angular frequency, equal to $2\pi/T$ for a T-periodic function
$f(c^{\pm})$	the left and right limits as t approaches c
$f'_{\pm}(c)$	the left and right derivatives at $t = c$
$L^1(D)$	space of absolutely integrable functions $f : D \to \mathbb{C}$
$L^2(D)$	space of square integrable functions $f : D \to \mathbb{C}$
$\hat{f}(\omega)$, $\mathcal{F}[f(t)]$	Fourier transform of $f \in L^1(\mathbb{R})$
$(f * g)(t)$	convolution of two functions $f, g : \mathbb{R} \to \mathbb{C}$
$F(s)$, $\mathcal{L}[f(t)]$	Laplace transform of $f : \mathbb{R}_{\geq 0} \to \mathbb{C}$
$\delta(t)$, $H(t)$	the Dirac delta function and Heaviside step function respectively
$A(z)$, $\mathcal{Z}[a]$	$\mathcal{Z}$-transform of a sequence $(a_n)_{n=0}^{\infty}$

1 Periodic functions and Fourier series

1.1 Complex valued functions

Throughout this course we will make extensive use of functions defined over the complex plane, particularly the *complex exponential function* e^z. Before diving into the heavy stuff, let us review the basics of complex functions and remind ourselves how the exponential function is extended from $\mathbb{R}$ to $\mathbb{C}$.

The field of complex numbers $\mathbb{C}$ consists of all numbers $z = x + \mathrm{i}y$, where x and y are real numbers and $\mathrm{i} = \sqrt{-1}$. Visualising $\mathbb{C}$ is easy; we simply draw the plane $\mathbb{R}^2$ and represent a complex number $x+\mathrm{i}y$ by the point $(x, y) \in \mathbb{R}^2$. We refer to this as the *complex plane* or *Argand plane*. Through this identification, the complex plane inherits a norm $|\cdot| : \mathbb{C} \to \mathbb{R}_{\geq 0}$ given by

$$|z| = |x + \mathrm{i}y| = \sqrt{x^2 + y^2}. \tag{1.1}$$

As with any good norm, it satisfies the *triangle inequality*

$$|z_1 + z_2| \leq |z_1| + |z_2|, \qquad \forall z_1, z_2 \in \mathbb{C}. \tag{1.2}$$

We often call $|z|$ the *modulus* of z. Associated to every complex number $z = x + \mathrm{i}y$ is its *complex conjugate* $\bar{z} = x - \mathrm{i}y$. Using this, we have the following formulae for the real and imaginary parts of z

$$\mathrm{Re}(z) = \frac{z + \bar{z}}{2}, \qquad \mathrm{Im}(z) = \frac{z - \bar{z}}{2\mathrm{i}}. \tag{1.3}$$

It is also easy to show that $|z|^2 = z\bar{z}$.

Now suppose we have a function $f : \mathbb{R} \to \mathbb{C}$. We write $f(t)$ as $u(t) + \mathrm{i}v(t)$ where $u(t)$ is the real part of $f(t)$ and $v(t)$ is the imaginary part. Doing so allows us to define the derivative of f as

$$f'(t) = u'(t) + \mathrm{i}v'(t). \tag{1.4}$$

Similarly, if $U(t)$ and $V(t)$ are anti-derivatives of $u(t)$ and $v(t)$ respectively, the integral of f over $[a, b]$ is given by

$$\int_a^b f(t)\,\mathrm{d}t = F(b) - F(a), \quad F(t) \doteq U(t) + \mathrm{i}V(t). \tag{1.5}$$

In other words, by splitting a complex valued function of a real variable into real and imaginary parts, we can do calculus as if we were dealing with two real valued functions.

Example 1.1. Many real functions can be extended to complex valued functions. For example, the exponential function e^x can be extended by noting that its Maclaurin series

$$\mathrm{e}^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}, \tag{1.6}$$

has an infinite radius of convergence and, therefore, is well-defined at any value of $z \in \mathbb{C}$.

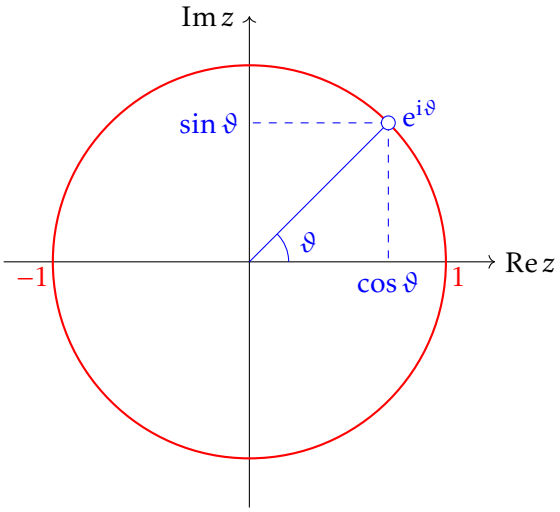


Figure 1: A plot of $z = e^{i\vartheta}$ in the complex plane.

Furthermore, if we set $z = i\vartheta$ with $\vartheta \in \mathbb{R}$, we can split $e^{i\vartheta}$ into real and imaginary parts as

$$\begin{aligned} e^{i\vartheta} = \sum_{n=0}^{\infty} \frac{(i\vartheta)^n}{n!} &= \left(1 - \frac{\vartheta^2}{2} + \frac{\vartheta^4}{24} - \dots\right) + i\left(\vartheta - \frac{\vartheta^3}{6} + \frac{\vartheta^5}{120} - \dots\right) \\ &= \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!}\vartheta^{2k} + i\sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!}\vartheta^{2k+1}. \end{aligned} \tag{1.7}$$

We now recognise the first sum as the Maclaurin series for $\cos\vartheta$ and the second as the Maclaurin series for $\sin\vartheta$. This gives the famous *Euler's formula*

$$e^{i\vartheta} = \cos\vartheta + i\sin\vartheta. \tag{1.8}$$

From this we can immediately deduce that $|e^{i\vartheta}| = 1$, so Euler's formula tells us that if we draw the function $f(\vartheta) = e^{i\vartheta}$ in the complex plane for $\vartheta \in \mathbb{R}$, we will get a circle of radius 1 (as shown in fig. 1).

We can derive several useful identities using Euler's formula. For example, using the fact that $\cos(-\vartheta) = \cos(\vartheta)$ and $\sin(-\vartheta) = -\sin(\vartheta)$, we see that

$$\overline{e^{i\vartheta}} = \cos\vartheta - i\sin\vartheta = \cos(-\vartheta) + i\sin(-\vartheta) = e^{-i\vartheta}. \tag{1.9}$$

Using the formulae for $\mathrm{Re}(z)$ and $\mathrm{Im}(z)$ together with Euler's formula, we obtain the following useful expressions for $\cos\vartheta$ and $\sin\vartheta$

$$\cos\vartheta = \frac{e^{i\vartheta} + e^{-i\vartheta}}{2}, \qquad \sin\vartheta = \frac{e^{i\vartheta} - e^{-i\vartheta}}{2i}. \tag{1.10}$$

As a sanity check, we can substitute these expressions into $\cos\vartheta + i\sin\vartheta$ and recover $e^{i\vartheta}$, as expected.

Example 1.2. (i) As Euler's formula gives us the real and imaginary parts of $e^{i\alpha\vartheta}$, we can calculate the integrals and derivatives as described above. Starting first with the derivative, we have

$$\frac{d}{d\vartheta}\left(e^{i\alpha\vartheta}\right) = -\alpha\sin\alpha\vartheta + i\alpha\cos\alpha\vartheta = i\alpha(\cos\alpha\vartheta + i\sin\alpha\vartheta) = i\alpha e^{i\alpha\vartheta}. \tag{1.11}$$

This also implies that $\frac{1}{i\alpha}e^{i\alpha\vartheta}$ is an anti-derivative of $e^{i\alpha\vartheta}$.

(ii) Now consider the function $e^{in\vartheta}$ with $n \in \mathbb{Z}$. First taking $n \neq 0$, we have

$$\int_0^{2\pi} e^{in\vartheta}\,d\vartheta = \left[\frac{e^{in\vartheta}}{in}\right]_0^{2\pi} = \frac{e^{2\pi in} - e^0}{in} = 0. \tag{1.12}$$

However, if $n = 0$, then $e^{in\vartheta} = 1$ and the integral is simply 2π. In summary

$$\int_0^{2\pi} e^{in\vartheta}\,d\vartheta = \begin{cases} 2\pi & n = 0, \\ 0 & n \neq 0. \end{cases} \tag{1.13}$$

1.2 Periodic functions and complex Fourier series

In many real life problems, we are interested in signals or functions that repeat after some time has elapsed. Electrical signals, audio waves and ocean waves provide just a few examples. Functions that display this property are called *periodic* and will be the focus of the rest of this chapter. Before getting to the interesting stuff, we have to formalise this concept.

Definition 1.1. A function $f : \mathbb{R} \to \mathbb{C}$ is *periodic with period* $T \neq 0$ if

$$f(t + T) = f(t), \tag{1.14}$$

for all $t \in \mathbb{R}$.

This definition implies that if f has period T, it also has period nT for any non-zero integer n, but we will often refer to the minimal period as 'the period of f'.

Example 1.3. We have the following examples of periodic functions:

- The functions $\sin t$ and $\cos t$ are 2π-periodic. By Euler's formula, e^{it} is also 2π-periodic.
- To obtain examples of a T-periodic function for arbitrary T, we can scale $t \mapsto 2\pi t/T$ in the above to get $\sin(2\pi t/T)$, $\cos(2\pi t/T)$ and $e^{2\pi it/T}$. More generally, if $f(t)$ is a T-periodic function, the scaled function $g(t) = f(tT/K)$ is K-periodic.
- There is no requirement that a periodic function needs to be continuous. For example,

we can consider the function $\chi_{\mathbb{Z}}(t)$ defined by

$$\chi_{\mathbb{Z}}(t) = \begin{cases} 1 & t \in \mathbb{Z}, \\ 0 & \text{otherwise.} \end{cases} \tag{1.15}$$

Concretely, it is equal to 1 when t is an integer and is 0 everywhere else. This function is discontinuous at every integer and has period 1. In fact, if you replace $\mathbb{Z}$ by $\mathbb{Q}$ in the above definition, you get a function that is everywhere discontinuous and has infinitely many periods of the form $1/p$ for every prime p.

As we'll see later, periodic functions have some nice properties, but a useful one for now is that integrating over a full period gives the same result regardless of the start point. That is, for any $a \in \mathbb{R}$

$$\int_a^{a+T} f(t)\,\mathrm{d}t = \int_0^T f(t)\,\mathrm{d}t, \tag{1.16}$$

if f has period T. To show this, you can split up the integral into two parts and change variables to shift the integration range to be of the form $[0, T]$.

Motivated by his work on the heat equation, Joseph Fourier asked if arbitrary periodic functions could be written as a series of simpler periodic functions. His claim was bold and somewhat imprecise, but the idea would prove to be revolutionary, leading to the development of *harmonic analysis*. This idea leads to the key definition of this chapter.

Definition 1.2. Let $f : \mathbb{R} \to \mathbb{C}$ be a T-periodic function. The (complex) *Fourier series* of f is given by

$$f(t) \sim \sum_{n\in\mathbb{Z}} c_n \mathrm{e}^{\mathrm{i}n\Omega t}, \qquad c_m = \frac{1}{T}\int_0^T f(t)\mathrm{e}^{-\mathrm{i}m\Omega t}\,\mathrm{d}t, \tag{1.17}$$

where $\Omega = 2\pi/T$ is the *angular frequency*. The numbers c_n are called the *Fourier coefficients*, while the finite series

$$S_N(t) = \sum_{n=-N}^{N} c_n \mathrm{e}^{\mathrm{i}n\Omega t}, \tag{1.18}$$

is called the Nth *partial sum* of the Fourier series.

To guarantee that these integrals exist, we demand that f must be *absolutely integrable* over a single period, meaning that

$$\int_0^T |f(t)|\,\mathrm{d}t < \infty. \tag{1.19}$$

This ensures that each c_n is well-defined since

$$\left|\int_0^T f(t)\mathrm{e}^{-\mathrm{i}n\Omega t}\,\mathrm{d}t\right| \le \int_0^T |f(t)\mathrm{e}^{-\mathrm{i}n\Omega t}|\,\mathrm{d}t = \int_0^T |f(t)|\,\mathrm{d}t, \tag{1.20}$$

therefore bounding the integral for any $n \in \mathbb{Z}$. It is easy to show that absolutely integrable functions form a vector space which we denote by $L^1([0,T])$.

As presented, this is just a definition of some series and we cannot really say how f relates to its Fourier series, so we use $\sim$ as shorthand for 'has Fourier series'. To motivate why this is at least a reasonable definition, let us suppose that $f(t)$ can be written as

$$f(t) = \sum_{n\in\mathbb{Z}} c_n e^{in\Omega t}. \tag{1.21}$$

Then we can multiply by $e^{-im\Omega t}$ and integrate to get

$$\int_0^T f(t)e^{-im\Omega t}\,dt = \sum_{n\in\mathbb{Z}}\left(\int_0^T c_n e^{i(n-m)\Omega t}\,dt\right), \tag{1.22}$$

assuming that we can exchange summation and integration. Now we change variables to $s = 2\pi t/T = \Omega t$ to change our integration range to $[0, 2\pi]$

$$\sum_{n\in\mathbb{Z}}\left(\int_0^T c_n e^{i(n-m)\Omega t}\,dt\right) = \frac{T}{2\pi}\sum_{n\in\mathbb{Z}}\left(\int_0^{2\pi} c_n e^{i(n-m)s}\,ds\right). \tag{1.23}$$

As seen in eq. (1.13), the only integral on the right-hand side that is non-zero is when $n-m=0$ where the integral is equal to 2π. Putting this together we have

$$\int_0^T f(t)e^{-im\Omega t}\,dt = Tc_m, \tag{1.24}$$

recovering the definition of the Fourier coefficients in the definition! So we see that the definition is at least consistent if $f(t)$ is equal to its Fourier series. We will return to the issue of when this is actually true later in this chapter, but for now we will turn a blind eye to it.

Example 1.4. The *sawtooth wave* is a 2π-periodic function defined on $[-\pi,\pi)$ by $f(t) = t$. For $n \neq 0$, the Fourier coefficients are given by

$$\begin{aligned} c_n &= \frac{1}{2\pi}\int_{-\pi}^{\pi} te^{-int}\,dt = \left[-\frac{te^{-int}}{2\pi in}\right]_{-\pi}^{\pi} + \frac{1}{2\pi in}\int_{-\pi}^{\pi} e^{-int}\,dt, \\ &= -\frac{1}{2\pi in}(\pi e^{-in\pi} + \pi e^{in\pi}), \\ &= \frac{(-1)^{n+1}}{in}, \end{aligned} \tag{1.25}$$

where we've used $e^{in\pi} = (-1)^n$ in the final line. Since this has a pole at $n=0$, we must check the $n=0$ case separately

$$c_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} t\,dt = 0. \tag{1.26}$$

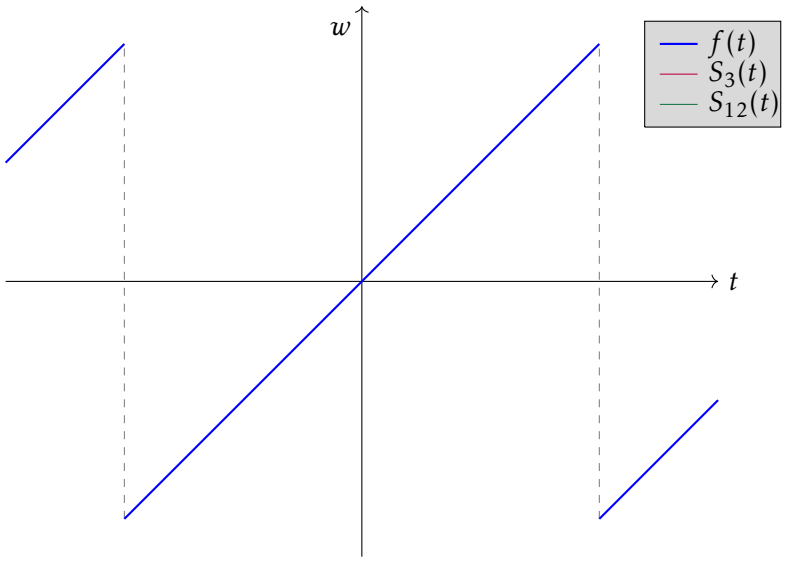


Figure 2: The $N = 3$ and $N = 12$ partial sums of the sawtooth wave compared with the function itself.

Therefore, $f(t)$ has Fourier series

$$f(t) \sim \sum_{n \neq 0} \frac{(-1)^{n+1}}{\mathrm{i}n} \mathrm{e}^{\mathrm{i}nt}. \tag{1.27}$$

Notice that the complex conjugate of this series is itself, showing that it is actually real. Plotting the partial sums for $N = 3$ and $N = 12$ gives fig. 2.

1.3 Trigonometric Fourier series

If we are interested in finding the Fourier series for a real function $f : \mathbb{R} \to \mathbb{R}$, it is often more convenient to try and write it in terms of $\cos(n\Omega t)$ and $\sin(n\Omega t)$. This can be done by using Euler's formula to rewrite the complex form as

$$f(t) \sim \sum_{n=-\infty}^{\infty} c_n \mathrm{e}^{\mathrm{i}\Omega nt} = \sum_{n=-\infty}^{\infty} c_n [\cos(n\Omega t) + \mathrm{i}\sin(n\Omega t)]. \tag{1.28}$$

Since $\cos(-t) = \cos(t)$ and $\sin(-t) = -\sin(t)$, we can pair up the terms in the sum with $n = k$ and $n = -k$ to get

$$\begin{aligned} f(t) &\sim c_0 + \sum_{n=1}^{\infty} [(c_n + c_{-n})\cos(n\Omega t) + \mathrm{i}(c_n - c_{-n})\sin(n\Omega t)], \\ &\doteq \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(n\Omega t) + b_n \sin(n\Omega t), \end{aligned} \tag{1.29}$$

where we've defined $a_n = c_n + c_{-n}$ and $b_n = \mathrm{i}(c_n - c_{-n})$. When f is a real function, it can be easily seen that $c_{-n} = \bar{c}_n$, so we can write $a_n = 2\mathrm{Re}(c_n)$ and $b_n = -2\mathrm{Im}(c_n)$. By substituting in the integral expression for the complex Fourier coefficients, we see that a_n and b_n can be computed via

$$a_n = \frac{2}{T}\int_0^T f(t)\cos(n\Omega t)\,\mathrm{d}t, \qquad b_n = \frac{2}{T}\int_0^T f(t)\sin(n\Omega t)\,\mathrm{d}t. \tag{1.30}$$

Note that the integral formula for a_n is consistent with the factor of $\frac{1}{2}$ multiplying a_0 in the expansion above.

Example 1.5. Returning to example 1.4, we saw that the function defined on $[-\pi, \pi)$ as $f(t) = t$ has complex Fourier coefficients

$$c_n = \begin{cases} \frac{(-1)^{n+1}}{\mathrm{i}n} & n \neq 0, \\ 0 & n = 0. \end{cases} \tag{1.31}$$

From this we see that $c_{-n} = -c_n$ for all n, so we have

$$a_n = c_n + c_{-n} = 0, \qquad b_n = \mathrm{i}(c_n - c_{-n}) = \frac{2(-1)^{n+1}}{n}. \tag{1.32}$$

The Fourier series of $f(t)$ in trigonometric form is therefore

$$f(t) \sim \sum_{n=1}^{\infty} \frac{2(-1)^{n+1}}{n} \sin nt. \tag{1.33}$$

Usually, it is easier to calculate the complex Fourier coefficients first and then use the linear relations to obtain the trigonometric coefficients. However, if f has certain symmetry properties a nice simplification occurs.

Definition 1.3. A function $f : \mathbb{R} \to \mathbb{C}$ is called *even* if

$$f(-x) = f(x), \quad \forall x \in \mathbb{R}. \tag{1.34}$$

Similarly, f is said to be *odd* if

$$f(-x) = -f(x), \quad \forall x \in \mathbb{R}. \tag{1.35}$$

There are several easy examples of odd and even functions. Functions of the form $f(t) = t^n$ are odd when n is odd and even when n is even, while $\cos t$ and $\sin t$ are even and odd respectively. However, note that $\mathrm{e}^{\mathrm{i}nt}$ is neither odd nor even for $n \neq 0$.

Clearly, the sum of two even (resp. odd) functions is also even (resp. odd), but under multiplication we must be more careful. While the product of two even functions is even, if f and g are odd functions, we have

$$f(-x)g(-x) = (-f(x))(-g(x)) = f(x)g(x), \tag{1.36}$$

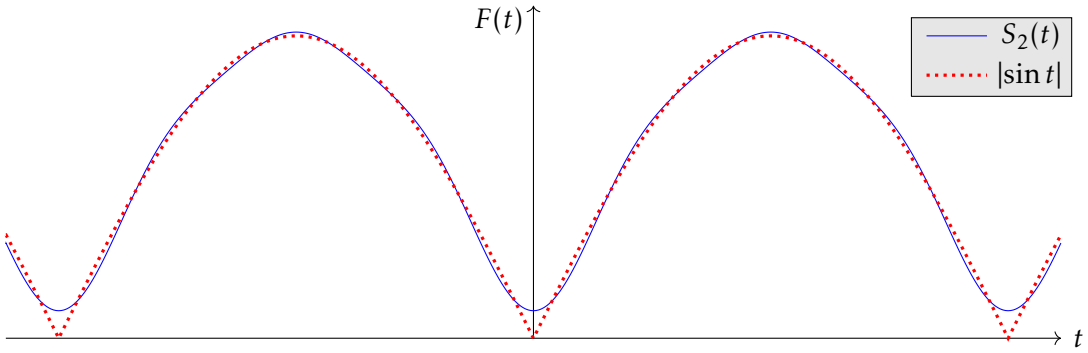


Figure 3: Fourier series with 3 terms for $f(t) = |\sin t|$.

meaning the product is actually even! It is also easy to show that the product of an odd function and an even function is always odd.

Due to the symmetry of odd and even functions under $t \mapsto -t$, integrals over the interval $[-a, a]$ simplify tremendously. Concretely, for any function f we have

$$\int_{-a}^{a} f(t)\,\mathrm{d}t = \int_{-a}^{0} f(t)\,\mathrm{d}t + \int_{0}^{a} f(t)\,\mathrm{d}t = \int_{0}^{a} (f(t) + f(-t))\,\mathrm{d}t, \tag{1.37}$$

where we've used the substitution $t \mapsto -t$ in the integral over $[-a, 0]$. Supposing now that f is either even or odd gives

$$\int_{-a}^{a} f(t)\,\mathrm{d}t = \begin{cases} 2\int_0^a f(t)\,\mathrm{d}t & f \text{ even}, \\ 0 & f \text{ odd}. \end{cases} \tag{1.38}$$

Using this formula, we see that if $f(t)$ is an even T-periodic function, then $f(t)\sin(n\Omega t)$ is odd and the sine Fourier coefficients b_n vanish. Similarly, if $f(t)$ is odd, then so is $f(t)\cos(n\Omega t)$, meaning a_n vanishes! In summary, we have

$$\begin{aligned} f \text{ even:} \quad & b_n = 0 \Longrightarrow f(t) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(n\Omega t), \quad a_n = \frac{4}{T}\int_0^{T/2} f(t)\cos(n\Omega t)\,\mathrm{d}t, \\ f \text{ odd:} \quad & a_n = 0 \Longrightarrow \quad f(t) \sim \sum_{n=1}^{\infty} b_n \sin(n\Omega t), \quad b_n = \frac{4}{T}\int_0^{T/2} f(t)\sin(n\Omega t)\,\mathrm{d}t. \end{aligned} \tag{1.39}$$

This is precisely why the Fourier series for the sawtooth wave only contains sine terms: it is an odd function, so $a_n = 0$ for every n. Furthermore, we can calculate b_n directly as

$$b_n = \frac{2}{\pi}\int_0^{\pi} t\sin nt\,\mathrm{d}t = \left[-\frac{2t\cos nt}{n\pi}\right]_0^{\pi} + \frac{2}{n\pi}\int_0^{\pi} \cos nt\,\mathrm{d}t = \frac{2(-1)^{n+1}}{n}, \tag{1.40}$$

matching what we got before.

Example 1.6. For a more involved example, consider the function $f(t) = |\sin t|$. This is an even function and π-periodic, so its Fourier series contains only terms of the form $\cos(2nt)$. The cosine Fourier coefficients are given by

$$a_n = \frac{4}{\pi}\int_0^{\pi/2} \sin(t)\cos(2nt)\,\mathrm{d}t. \tag{1.41}$$

We can use the product-to-sum identities to get

$$a_n = \frac{2}{\pi}\int_0^{\pi/2} [\sin(1+2n)t + \sin(1-2n)t]\,\mathrm{d}t = \frac{4}{\pi(1-4n^2)}. \tag{1.42}$$

Note that since $n = 0$ is not a pole of a_n, we do not need to evaluate a_0 separately. This gives us the Fourier expansion

$$f(t) \sim \frac{2}{\pi} - \frac{4}{\pi}\sum_{n=1}^{\infty} \frac{\cos(2nt)}{4n^2-1}. \tag{1.43}$$

A comparison between $f(t)$ and the first three terms of the Fourier series is shown in fig. 3.

1.4 Half-range series

Quite often it is useful to obtain a Fourier series for a function $f(t)$ that is only defined on an interval $[0, T/2)$. This often occurs when solving PDEs, for example. In order to do this, we need to extend $f(t)$ to a periodic function over $\mathbb{R}$, but there are several ways to do this. Since we have the freedom to extend it however we like, we can choose the extension to be either odd or even and the Fourier series computation becomes fairly straightforward. Given such an f, we define the *even extension* $F_{\text{even}}(t)$ to be

$$F_{\text{even}}(t) = \begin{cases} f(t) & \text{if } 0 \le t < T/2, \\ f(-t) & \text{if } -T/2 < t < 0. \end{cases} \tag{1.44}$$

Treating this as an even T-periodic function, we can Fourier expand it as in the previous section

$$F_{\text{even}}(t) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(n\Omega t), \quad a_n = \frac{4}{T}\int_0^{T/2} f(t)\cos(n\Omega t)\,\mathrm{d}t, \tag{1.45}$$

where $\Omega = 2\pi/T$. We call this the *Fourier cosine series* or *half-range cosine series*. Similarly, the *odd extension* of f is defined as

$$F_{\text{odd}}(t) = \begin{cases} f(t) & \text{if } 0 \le t < T/2, \\ -f(-t) & \text{if } -T/2 < t < 0, \end{cases} \tag{1.46}$$

which leads to the *Fourier sine series* or *half-range sine series*

$$F_{\text{odd}}(t) \sim \sum_{n=1}^{\infty} b_n \sin(n\Omega t), \quad b_n = \frac{4}{T}\int_0^{T/2} f(t)\sin(n\Omega t)\,\mathrm{d}t. \tag{1.47}$$

On the interval $[0, T/2)$, where the original function was defined, these two series should agree provided they converge. So we are free to choose which one to work with if we require a half-range series.

Example 1.7. Consider the function $f(t) = t$ on the half-range $[0, \pi)$. We can extend this to an even function and calculate its cosine Fourier coefficients as

$$\begin{aligned} a_n &= \frac{2}{\pi}\int_0^\pi t\cos nt\,\mathrm{d}t = \frac{2}{\pi}\left[\frac{t\sin nt}{n}\right]_0^\pi - \frac{2}{\pi n}\int_0^\pi \sin nt\,\mathrm{d}t, \\ &= \frac{2}{\pi n^2}[\cos nt]_0^\pi, \\ &= \frac{2((-1)^n - 1)}{\pi n^2}, \end{aligned} \tag{1.48}$$

so a_n is non-zero only when n is odd. It is easy to see that $a_0 = \pi$, so the cosine Fourier series is therefore

$$F_{\text{even}}(t) \sim \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2((-1)^n - 1)}{\pi n^2}\cos nt = \frac{\pi}{2} - \sum_{m=0}^{\infty} \frac{4}{\pi(2m+1)^2}\cos(2m+1)t, \tag{1.49}$$

where in the final equality we've introduced $n = 2m + 1$.

1.5 Decay rate of Fourier coefficients

So far we've been calculating Fourier series without really knowing if the series actually captures the original function in question, using $\sim$ to just mean 'has Fourier series'. We now turn to this question, first justifying when these series have a finite value.

Recall from single variable analysis that infinite series can display different behaviours. For example, we have the following.

- The harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent as the partial sums do not tend to a finite limit;
- The alternating harmonic series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ is *conditionally convergent*, meaning that it has a finite limit, but the sum of the absolute values does not converge;
- Finally, the sum $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$ is *absolutely convergent*, meaning both it and the sum of absolute values converge to a finite value.

Our goal is to establish how the Fourier coefficients decay as $n \to \infty$ to see when the Fourier partial sums converge.

Proposition 1.1. Let f be an absolutely integrable T-periodic function with Fourier coefficients c_n. Then

$$\text{(i) } |c_n| \leq \frac{1}{T}\int_0^T |f(t)|\,\mathrm{d}t, \qquad \text{(ii) } \lim_{n\to\pm\infty} c_n = 0.$$

Proof. (i) The triangle inequality for integrals implies that

$$|c_n| \leq \left| \frac{1}{T} \int_0^T f(t) \mathrm{e}^{-\mathrm{i}n\Omega t} \, \mathrm{d}t \right| \leq \frac{1}{T} \int_0^T |f(t) \mathrm{e}^{-\mathrm{i}n\Omega t}| \, \mathrm{d}t = \frac{1}{T} \int_0^T |f(t)| \, \mathrm{d}t, \tag{1.50}$$

which is finite since f is absolutely integrable.

(ii) This is known as the Riemann-Lebesgue lemma and proving it is beyond the scope of the course. □

This proposition tells us that the Fourier coefficients are bounded above and tend to zero as $n \to \infty$, but this isn't enough for convergence. For example, every term of the divergent harmonic series is bounded by 1 and approaches zero as n grows, so we need a stronger result to ensure convergence. The natural next step is to impose more structure on f.

Proposition 1.2. Let f be a continuous T-periodic function that is k-times continuously differentiable with $k \geq 1$. Then there exists an $M > 0$ such that

$$|c_n| \leq \frac{M}{|n|^k}, \quad \forall n \neq 0, \tag{1.51}$$

where c_n are the Fourier coefficients of f.

Proof. By hypothesis $f^{(k)}$ is T-periodic and continuous, so $f^{(k)} \in L^1([0, T])$. We can therefore Fourier expand $f^{(k)}$ as

$$f^{(k)}(t) \sim \sum_{n \in \mathbb{Z}} d_n \mathrm{e}^{\mathrm{i}n\Omega t}, \qquad d_n = \frac{1}{T} \int_0^T f^{(k)}(t) \mathrm{e}^{-\mathrm{i}n\Omega t} \, \mathrm{d}t. \tag{1.52}$$

We can integrate this by parts to obtain

$$\begin{aligned} d_n &= \frac{1}{T} \left[f^{(k-1)}(t) \mathrm{e}^{-\mathrm{i}n\Omega t} \right]_0^T + \frac{1}{T} \int_0^T f^{(k-1)}(t) (\mathrm{i}\Omega n) \mathrm{e}^{-\mathrm{i}n\Omega t} \, \mathrm{d}t, \\ &= \frac{1}{T} (f^{(k-1)}(T) - f^{(k-1)}(0)) + \frac{\mathrm{i}\Omega n}{T} \int_0^T f^{(k-1)}(t) \mathrm{e}^{-\mathrm{i}n\Omega t} \, \mathrm{d}t, \\ &= \frac{\mathrm{i}\Omega n}{T} \int_0^T f^{(k-1)}(t) \mathrm{e}^{-\mathrm{i}n\Omega t} \, \mathrm{d}t \end{aligned} \tag{1.53}$$

where the boundary term vanished as $f^{(k-1)}$ is T-periodic. We can repeat this process until we reach

$$d_n = \frac{(\mathrm{i}\Omega n)^k}{T} \int_0^T f(t) \mathrm{e}^{-\mathrm{i}n\Omega t} \, \mathrm{d}t = (\mathrm{i}\Omega n)^k c_n. \tag{1.54}$$

Using part (i) of the previous proposition, we can bound $|d_n|$ by $\frac{1}{T} \int_0^T |f^{(k)}(t)| \, \mathrm{d}t$, which in turn allows us to bound $|c_n|$ as

$$|c_n| = \frac{|d_n|}{|\mathrm{i}\Omega n|^k} \leq \frac{1}{T(\Omega |n|)^k} \int_0^T |f^{(k)}(t)| \, \mathrm{d}t \doteq \frac{M}{|n|^k}, \tag{1.55}$$

giving the result. □

Now consider the case where f is a T-periodic function that is twice continuously differentiable. Then by proposition 1.2, we have

$$\left|\sum_{n\in\mathbb{Z}} c_n e^{i\Omega nt}\right| \leq |c_0| + \sum_{n=1}^{\infty}(|c_n| + |c_{-n}|) \leq |c_0| + 2M\sum_{n=1}^{\infty}\frac{1}{n^2}, \tag{1.56}$$

where $M = \frac{1}{T\Omega^2}\int_0^T |f''(t)|\,\mathrm{d}t$. As remarked earlier, the series $\sum_{n=1}^{\infty}\frac{1}{n^2}$ is convergent, so we conclude that the right-hand side is finite and the Fourier series for $f(t)$ is absolutely convergent.

1.6 Convergence of Fourier series

The previous section shows that under certain assumptions on the smoothness of f, the partial sums of a Fourier series converge absolutely to a finite number with a particular decay rate. However, if we relax the assumption of continuity, we don't necessarily have a divergent series; it could well be conditionally convergent. With this in mind, we now compare how the limit of the partial sums relates to the value of f at that point, with particular care taken at points of discontinuity.

Recall that the left and right limits of a function are defined to be

$$f(c^-) \doteq \lim_{t\to c^-} f(t), \qquad f(c^+) \doteq \lim_{t\to c^+} f(t), \tag{1.57}$$

respectively. That is, the left limit is given by approaching $t = c$ from the left, keeping $t < c$ as we approach it, while the right limit keeps $t > c$. For a continuous function, these two must match

$$f(c) = f(c^+) = f(c^-), \tag{1.58}$$

but for discontinuous functions they don't. In particular, if we plot a discontinuous function, we will see a jump of size $|f(c^+) - f(c^-)|$ at $t = c$. We can similarly define the left and right derivatives at a point, regardless of whether f is continuous. The *left derivative* at $t = c$ is given by

$$f'_-(c) = \lim_{h\to 0^-}\frac{f(c+h) - f(c^-)}{h}, \tag{1.59}$$

while the *right derivative* is defined as

$$f'_+(c) = \lim_{h\to 0^+}\frac{f(c+h) - f(c^+)}{h}. \tag{1.60}$$

Again, if f is continuous and differentiable, these two limits agree and we can unambiguously define $f'(c)$.

Example 1.8. For any $a \in \mathbb{R}$, consider the function

$$f(t) = \begin{cases} t^2, & t < 0, \\ t - a, & t \geq 0. \end{cases} \tag{1.61}$$

This is discontinuous at $t = 0$ if $a \neq 0$. Indeed, approaching from the left the limit is

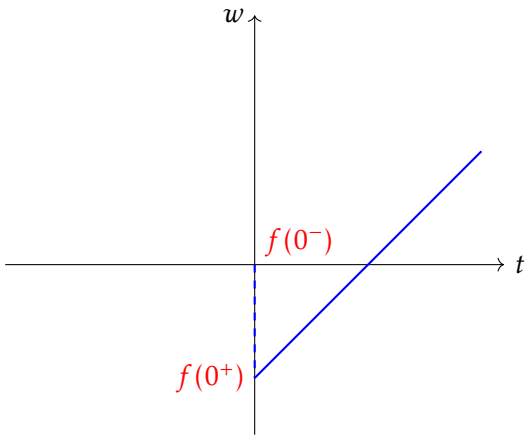


Figure 4: The function given in eq. (1.61) drawn with $a > 0$.

$f(0^-) = \lim_{t\to 0^-} t^2 = 0$, while the right limit is $f(0^+) = \lim_{t\to 0^+}(t-a) = -a$. The left and right derivatives are given by

$$f'_-(0) = \lim_{h\to 0^-} \frac{h^2 - 0}{h} = \lim_{h\to 0^-} h = 0, \tag{1.62}$$

and

$$f'_+(0) = \lim_{h\to 0^+} \frac{(h-a)+a}{h} = \lim_{h\to 0^+} \frac{h}{h} = 1. \tag{1.63}$$

Therefore, $f(t)$ is not differentiable at $t = 0$ for any value of a.

Equipped with this, we can now state *Dirichlet's criterion* for pointwise convergence of Fourier series.

Theorem 1.1. Let f be a T-periodic function and assume that $f'_+(t)$ and $f'_-(t)$ exist everywhere. Then the Fourier partial sums $S_N(t)$ converge pointwise to

$$\lim_{N\to\infty} S_N(t) = \frac{f(t^+) + f(t^-)}{2} = \lim_{\varepsilon\to 0^+} \frac{f(t+\varepsilon) + f(t-\varepsilon)}{2}, \tag{1.64}$$

for every $t \in \mathbb{R}$.

Remark. Note that Dirichlet's criterion has many different forms, so you might see various different formulations in other texts. A popular form in engineering textbooks is to require that f be integrable and have finitely many discontinuities and local extrema over a single period.

Given a function that satisfies the Dirichlet criterion, we can now replace the symbol $\sim$ with a genuine equality at points of continuity. On the other hand, at points of discontinuity the partial sums tend to the average of the left and right limits. We've actually already seen this in

action with the sawtooth wave. There, the Fourier series takes the form

$$f(t) \sim \sum_{n=1}^{\infty} \frac{2(-1)^{n+1}}{n} \sin nt, \tag{1.65}$$

with $t = \pi$ being a point of discontinuity. Substituting this into the Fourier series we get exactly zero since $\sin n\pi = 0$ for $n \in \mathbb{Z}$. Comparing this with the left and right limits, we see

$$f(\pi^-) = \pi, \quad f(\pi^+) = -\pi \implies \frac{f(\pi^+) + f(\pi^-)}{2} = 0, \tag{1.66}$$

matching the expectation from the Dirichlet criterion. Indeed, it is easy to see that the left and right derivatives are well defined at $t = \pi$, justifying the equality.

Example 1.9. Let's return to $f(t) = |\sin t|$ and investigate Fourier series convergence at $t = 0$. First note that $f(t)$ is continuous at $t = 0$, but not differentiable. Nonetheless, the left and right derivatives are well-defined and given by

$$f'_-(0) = \lim_{h \to 0^-} \frac{-\sin h}{h} = -1, \quad f'_+(0) = \lim_{h \to 0^+} \frac{\sin h}{h} = 1. \tag{1.67}$$

The final equalities can be justified by using the Maclaurin series for $\sin h$ or by using l'Hôpital's rule. Therefore, the Fourier series for $f(t)$ converges everywhere to $f(t)$ and we can write

$$f(t) = \frac{2}{\pi} - \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{\cos(2nt)}{4n^2 - 1}. \tag{1.68}$$

Taking $t = 0$ then gives us

$$0 = \frac{2}{\pi} - \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{4n^2 - 1} \implies \sum_{n=1}^{\infty} \frac{1}{4n^2 - 1} = \frac{1}{2}. \tag{1.69}$$

1.7 L^2 theory and Parseval's theorem

So far, we've neglected an interesting property of periodic functions. Namely, if f and g are both T-periodic functions, then $\alpha f + \beta g$ is also T-periodic for any choice of constants α and β. This gives the space of T-periodic functions the structure of a *vector space* where the functions take on the role of 'vectors'. This begs the question, what other notions from linear algebra can we use to study T-periodic functions?

First, recall that for usual finite length vectors in $\mathbb{C}^n$, we have the standard inner product

$$\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^{n} u_i \overline{v}_i. \tag{1.70}$$

This inner product encodes important geometric data about a vector. For example, the squared length of a vector is given by

$$|\mathbf{u}|^2 = \mathbf{u} \cdot \mathbf{u} = \sum_{i=1}^{n} |u_i|^2. \tag{1.71}$$

Furthermore, it satisfies the *Cauchy-Schwarz inequality*

$$|\mathbf{u}\cdot\mathbf{v}| \leq |\mathbf{u}||\mathbf{v}|. \tag{1.72}$$

When two vectors satisfy $\mathbf{u}\cdot\mathbf{v}=0$, we say that $\mathbf{u}$ and $\mathbf{v}$ are *orthogonal*, extending the idea from real vector spaces that the two vectors lie on perpendicular lines through the origin. This is useful, as we can define an *orthonormal (ON) basis* to be a set of vectors $\{\mathbf{u}_i : i=1,\dots,n\}$ that satisfy

$$\mathbf{u}_i\cdot\mathbf{u}_j = \begin{cases} 1, & \text{if } i=j, \\ 0, & \text{if } i\neq j. \end{cases} \tag{1.73}$$

In doing so, we can expand any vector in $\mathbb{C}^n$ in terms of this basis

$$\mathbf{v} = \sum_{i=1}^{n} \alpha_i \mathbf{u}_i, \tag{1.74}$$

from which we want to calculate the constants α_i. Since the $\{\mathbf{u}_i\}$ form an orthonormal basis, we can take the inner product of $\mathbf{v}$ with $\mathbf{u}_m$ for some m to get

$$\mathbf{u}_m\cdot\mathbf{v} = \sum_{i=1}^{n} \alpha_i(\mathbf{u}_m\cdot\mathbf{u}_i) = \alpha_m, \tag{1.75}$$

where we've used the fact that $\mathbf{u}_m\cdot\mathbf{u}_i$ is only non-zero when $m=i$. So any vector can be easily expanded into an ON basis by calculating its inner products with the basis vectors.

This all seems fairly similar to Fourier series: we have some functions that we want to write as sums of simpler functions like $\mathrm{e}^{\mathrm{i}n\Omega t}$ and then we calculate the coefficients of this expansion by integrating $f(t)$ against these simple functions. We can make this more concrete by defining the inner product of two T-periodic functions to be

$$\langle f,g\rangle = \frac{1}{T}\int_0^T f(t)\overline{g(t)}\,\mathrm{d}t. \tag{1.76}$$

This is a natural extension of the inner product for vectors, now we just think about our 'vectors' as being indexed by the real numbers in $[0,T)$ and the summation is replaced by integration. If we now restrict to functions satisfying

$$\|f\|^2 \doteq \langle f,f\rangle = \frac{1}{T}\int_0^T |f(t)|^2\,\mathrm{d}t < \infty, \tag{1.77}$$

we obtain the so-called *Lebesgue space* $L^2([0,T])$. Equipping this space with the above inner product makes it into an *inner product space*. It is easy to check that

$$\langle \mathrm{e}^{\mathrm{i}m\Omega t}, \mathrm{e}^{\mathrm{i}n\Omega t}\rangle = \begin{cases} 1, & \text{if } m=n, \\ 0, & \text{if } m\neq n. \end{cases} \tag{1.78}$$

This signals that the functions $f_m(t) = e^{im\Omega t}$ act like an ON basis for $L^2([0,T])$, though mathematically we should be a little more careful. We haven't established that it is a complete basis and haven't dealt with the nuances of having an infinite set of basis vectors, but we will be cavalier and assume things are fine. The bottom line is that expanding a T-periodic function that belongs to $L^2([0,T])$ into complex exponentials is essentially just an ON basis expansion where the Fourier coefficients are given by

$$c_m \doteq \langle f(t), e^{im\Omega t}\rangle = \frac{1}{T}\int_0^T f(t)e^{-im\Omega t}\,dt. \tag{1.79}$$

Similar statements hold for the trigonometric expansions, it is just a basis change from the exponential basis.

The quantity $\|f\|^2$ is the analogue of the length of a vector. In that case, instead of measuring the length directly, we expand into an ON basis and then use Pythagoras's theorem. We can do the same in $L^2([0,T])$, it goes by the name of *Parseval's theorem.*

Theorem 1.2. If $f \in L^2([0,T])$ has Fourier series

$$f(t) \sim \sum_{n\in\mathbb{Z}} c_n e^{in\Omega t} = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n\cos(n\Omega t) + b_n\sin(n\Omega t), \tag{1.80}$$

then

$$\frac{1}{T}\int_0^T |f(t)|^2\,dt = \sum_{n\in\mathbb{Z}}|c_n|^2 = \frac{|a_0|^2}{4} + \frac{1}{2}\sum_{n=1}^{\infty}\left(|a_n|^2 + |b_n|^2\right). \tag{1.81}$$

Proof. Substituting the Fourier expansion of $\overline{f(t)}$ into $\|f\|^2$ gives

$$\begin{aligned}\|f\|^2 &= \frac{1}{T}\int_0^T f(t)\left(\sum_{n\in\mathbb{Z}}\overline{c}_n e^{-in\Omega t}\right)dt,\\ &= \sum_{n\in\mathbb{Z}}\frac{\overline{c}_n}{T}\int_0^T f(t)e^{-in\Omega t}\,dt,\end{aligned} \tag{1.82}$$

where we've assumed that we can switch integration and summation. We now recognise the integral as the definition of $c_m = \langle f(t), e^{im\Omega t}\rangle$, so the sum becomes

$$\|f\|^2 = \sum_{n\in\mathbb{Z}}\overline{c}_n c_n = \sum_{n\in\mathbb{Z}}|c_n|^2. \tag{1.83}$$

Rewriting the c_n in terms of a_n and b_n and expanding the squares then gives the trigonometric result. □

Example 1.10. The sawtooth function belongs to $L^2([0,T])$, so we can use Parseval's theorem.

On one side we have

$$\|f\|^2 = \frac{1}{2\pi}\int_{-\pi}^{\pi} t^2\,\mathrm{d}t = \frac{1}{2\pi}\left[\frac{t^3}{3}\right]_{-\pi}^{\pi} = \frac{\pi^2}{3}. \tag{1.84}$$

The Fourier series is given by eq. (1.27), giving the sum

$$\sum_{n\in\mathbb{Z}} |c_n|^2 = \sum_{n\neq 0}\frac{1}{n^2} = 2\sum_{n=1}^{\infty}\frac{1}{n^2}. \tag{1.85}$$

Equating both results and dividing by two then gives

$$\sum_{n=1}^{\infty}\frac{1}{n^2} = \frac{\pi^2}{6}. \tag{1.86}$$

This is the famous solution to the Basel problem, first solved by Euler in 1734.

1.8 The Gibbs phenomenon ★

— Work in progress, very non-examinable —

2 The Fourier transform

2.1 Definition and properties

Fourier series allow us to view periodic functions in a different light. In particular, they break up a function into sines and cosines (or complex exponentials), each of which has a frequency a multiple of $\Omega = 2\pi/T$, with the Fourier coefficients telling us how to weight each of these in the sum. In this sense, we can think of the sequence of Fourier coefficients as providing a different representation of the function, one in terms of the frequency instead of t. So what happens if we try to do this for non-periodic functions by taking $T \to \infty$?

Heuristically, by taking $T \to \infty$, the fundamental frequency Ω tends to zero and we must take into account all possible frequencies $\omega \in \mathbb{R}$. This is telling us that the analogue of the Fourier series for a non-periodic function must be an integral instead of a sum. This is the basis of the *Fourier transform* and the frequency representation of f, which we aim to motivate in this chapter.

First we want to generalise the idea of the Fourier coefficients, but now that we're dealing with functions that aren't periodic, we need to restrict to functions that are suitably well behaved over all of $\mathbb{R}$. This leads to the following definition.

Definition 2.1. The *Lebesgue space* $L^1(\mathbb{R})$ consists of all functions $f : \mathbb{R} \to \mathbb{C}$ such that

$$\int_{-\infty}^{\infty} |f(t)| \, \mathrm{d}t < \infty. \tag{2.1}$$

Suppose two functions f and g belong to $L^1(\mathbb{R})$. Then for any complex numbers α and β we have

$$\int_{-\infty}^{\infty} |\alpha f(t) + \beta g(t)| \, \mathrm{d}t \leq |\alpha| \int_{-\infty}^{\infty} |f(t)| \, \mathrm{d}t + |\beta| \int_{-\infty}^{\infty} |g(t)| \, \mathrm{d}t, \tag{2.2}$$

which is finite since $f, g \in L^1(\mathbb{R})$. Therefore, $L^1(\mathbb{R})$ is a vector space.

Definition 2.2. Let $f \in L^1(\mathbb{R})$. The *Fourier transform* of f is given by

$$\hat{f}(\omega) = \mathcal{F}[f(t)] = \int_{-\infty}^{\infty} f(t) \mathrm{e}^{-\mathrm{i}\omega t} \, \mathrm{d}t. \tag{2.3}$$

The Fourier transform generalises the Fourier coefficients in a natural way, extending the integration range to the whole real line and taking into account all possible frequencies ω. We can think of it as taking a function of time t and repackaging it into a function of frequency ω instead, so it is reasonable to say 'the Fourier transform lives in the frequency domain'. If it can be established that we can reconstruct $f(t)$ from $\hat{f}(\omega)$, then $\hat{f}(\omega)$ encodes all the relevant information about $f(t)$ in ways that might be more useful for certain problems. We will see this later, but for now let's compute an example.

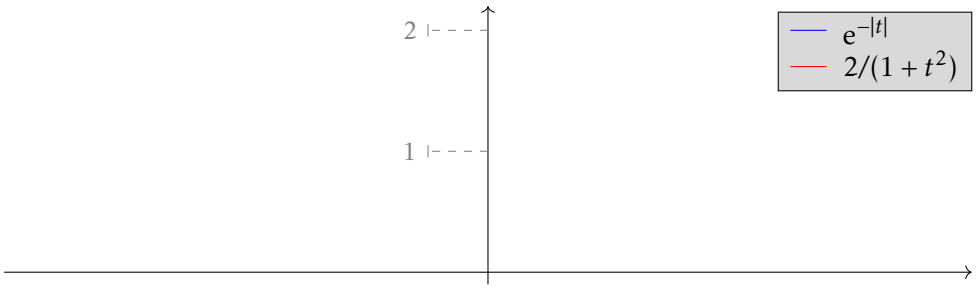


Figure 5: The function $f(t) = e^{-|t|}$ compared with its Fourier transform $\hat{f}(\omega) = 2/(1+\omega^2)$.

Example 2.1. Consider the function $f(t) = e^{-|t|}$. This belongs to $L^1(\mathbb{R})$ because

$$\int_{-\infty}^{\infty} |e^{-|t|}| \, dt = 2 \int_0^{\infty} e^{-t} \, dt = 2, \tag{2.4}$$

so we can calculate its Fourier transform. In particular, by splitting the integral range into $(-\infty, 0] \cup [0, \infty)$ we have

$$\begin{aligned} \hat{f}(\omega) = \int_{-\infty}^{\infty} e^{-|t| - i\omega t} \, dt &= \int_{-\infty}^{0} e^{(1-i\omega)t} \, dt + \int_0^{\infty} e^{-(1+i\omega)t} \, dt, \\ &= \left[\frac{e^{(1-i\omega)t}}{1-i\omega} \right]_{-\infty}^{0} + \left[-\frac{e^{-(1+i\omega)t}}{1+i\omega} \right]_0^{\infty}. \end{aligned} \tag{2.5}$$

Now note that $e^{(1-i\omega)t}$ vanishes as $t \to -\infty$ since the e^{-t} factor suppresses it. Likewise, $e^{-(1+i\omega)t}$ vanishes as $t \to \infty$, so we get

$$\hat{f}(\omega) = \frac{1}{1-i\omega} + \frac{1}{1+i\omega} = \frac{2}{1+\omega^2}. \tag{2.6}$$

Both f and $\hat{f}$ are plotted in fig. 5.

In this example, the resulting Fourier transform $\hat{f}(\omega) = 2/(1+\omega^2)$ also belongs to $L^1(\mathbb{R})$. This is not always the case! Consider the indicator function

$$\chi_{[-a,a]}(t) = \begin{cases} 1, & -a < t < a, \\ 0, & \text{otherwise.} \end{cases} \tag{2.7}$$

It is easy to show that the Fourier transform is given by

$$\hat{\chi}_{[-a,a]}(\omega) = \frac{2 \sin a\omega}{\omega}. \tag{2.8}$$

This function does not belong to $L^1(\mathbb{R})$ because the tails fall off as $1/|\omega|$ as $|\omega| \to \infty$. In general, the Fourier transform of an $L^1(\mathbb{R})$ function is a continuous function that vanishes at $\pm\infty$, due to the Riemann-Lebesgue lemma.

function	Fourier transform
$f(t)$	$\hat{f}(\omega)$
$\alpha f(t) + \beta g(t)$	$\alpha\hat{f}(\omega) + \beta\hat{g}(\omega)$
$f(at)$	$\frac{1}{\lvert a\rvert}\hat{f}\left(\frac{\omega}{a}\right)$
$f(t+b)$	$\mathrm{e}^{\mathrm{i}\omega b}\hat{f}(\omega)$
$tf(t)$	$\mathrm{i}\frac{\mathrm{d}\hat{f}}{\mathrm{d}\omega}$
$f'(t)$	$\mathrm{i}\omega\hat{f}(\omega)$
$\hat{f}(t)$	$2\pi f(-\omega)$
$(f * g)(t)$	$\hat{f}(\omega)\hat{g}(\omega)$

Table 1: Properties of the Fourier transform. Here α, β, a and b are constants with $a \neq 0$.

2.2 Transformation properties

Suppose we are given a function $f(t)$ and its Fourier transform $\hat{f}(\omega)$, but we want to find the Fourier transform of $g(t) = f(at+b)$ for $a \neq 0$. Instead of evaluating the integral directly, we can use the substitution rules for integrals to write $\hat{g}(\omega)$ in terms of $\hat{f}(\omega)$. Concretely, $\hat{g}(\omega)$ is given by

$$\hat{g}(\omega) = \mathcal{F}[f(at+b)] = \int_{-\infty}^{\infty} f(at+b)\mathrm{e}^{-\mathrm{i}\omega t}\,\mathrm{d}t. \tag{2.9}$$

Introducing the variable $u = at + b$, so $t = (u-b)/a$, we can rewrite the integral as

$$\hat{g}(\omega) = \int_{-\infty}^{\infty} f(u)\mathrm{e}^{-\mathrm{i}\omega(u-b)/a}\frac{\mathrm{d}u}{|a|} = \frac{\mathrm{e}^{\mathrm{i}\omega b/a}}{|a|}\hat{f}\left(\frac{\omega}{a}\right). \tag{2.10}$$

To justify why $|a|$ appears instead of a, note that if $a < 0$ the integration limits reverse to $(\infty, -\infty)$. However, we can switch the order of the limits by introducing a minus sign, taking the $1/a$ to $1/|a|$.

Now suppose we want to find the Fourier transform of $f'(t)$ instead. This can be done via integration by parts

$$\begin{aligned}\mathcal{F}[f'(t)] &= \int_{-\infty}^{\infty} f'(t)\mathrm{e}^{-\mathrm{i}\omega t}\,\mathrm{d}t, \\ &= \left[f(t)\mathrm{e}^{-\mathrm{i}\omega t}\right]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} \mathrm{i}\omega f(t)\mathrm{e}^{-\mathrm{i}\omega t}\,\mathrm{d}t.\end{aligned} \tag{2.11}$$

Since we've assumed $f(t) \in L^1(\mathbb{R})$, we must have that $f(t)$ vanishes at $\pm\infty$, so the boundary terms vanish. But now we're just left with the integral, which we recognise as

$$\mathcal{F}[f'(t)] = \mathrm{i}\omega\hat{f}(\omega). \tag{2.12}$$

This is perhaps the most important property of the Fourier transform: it turns derivatives by t into multiplication by $\mathrm{i}\omega$ in the frequency domain, turning highly difficult differential problems

into reasonably basic equations for $\hat{f}(\omega)$. This idea also holds in the opposite direction, taking the derivative of the Fourier transform formula gives

$$\frac{\mathrm{d}\hat{f}}{\mathrm{d}\omega} = \int_{-\infty}^{\infty} \frac{\mathrm{d}}{\mathrm{d}\omega}\left(f(t)\mathrm{e}^{-\mathrm{i}\omega t}\right)\mathrm{d}t = -\mathrm{i}\int_{-\infty}^{\infty} tf(t)\mathrm{e}^{-\mathrm{i}\omega t}\,\mathrm{d}t, \tag{2.13}$$

where we're allowed to move the derivative into the integral since $f \in L^1(\mathbb{R})$. The integral is now just the Fourier transform of $tf(t)$, so we have

$$\mathcal{F}[tf(t)] = \mathrm{i}\frac{\mathrm{d}\hat{f}}{\mathrm{d}\omega}, \tag{2.14}$$

thus turning multiplication by t into differentiation in the frequency domain.

Example 2.2. (i) We will find the Fourier transform of the Gaussian function $f(t) = \mathrm{e}^{-t^2/2}$. The integral expression for $\hat{f}(\omega)$ is given by

$$\hat{f}(\omega) = \int_{-\infty}^{\infty} \mathrm{e}^{-t^2/2}\mathrm{e}^{-\mathrm{i}\omega t}\,\mathrm{d}t, \tag{2.15}$$

but this can be a little tricky to evaluate. Instead, we can indirectly find $\hat{f}$ by noting that

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\mathrm{e}^{-t^2/2}\right) = -t\mathrm{e}^{-t^2/2} = -tf(t) \tag{2.16}$$

Using the rules for Fourier transforms of derivatives and $tf(t)$ we obtain the differential equation

$$\frac{\mathrm{d}\hat{f}}{\mathrm{d}\omega} = -\omega\hat{f}(\omega). \tag{2.17}$$

Rearranging this, we get $\hat{f}'(\omega)/\hat{f}(\omega) = -\omega$ and recognise the left-hand side as the derivative of $\log\hat{f}(\omega)$. Integrating this rearrangement, we arrive at

$$\log\hat{f}(\omega) = -\frac{\omega^2}{2} + \log C \Longrightarrow \hat{f}(\omega) = C\mathrm{e}^{-\omega^2/2}, \tag{2.18}$$

where $\log C$ is the constant of integration. We've therefore determined, up to a constant, that the Fourier transform of a Gaussian is another Gaussian. By using the fact that $\hat{f}(0) = \int_{-\infty}^{\infty}\mathrm{e}^{-t^2/2}\,\mathrm{d}t = \sqrt{2\pi}$, we see that $C = \sqrt{2\pi}$. In summary, we have

$$\mathcal{F}[\mathrm{e}^{-t^2/2}] = \sqrt{2\pi}\mathrm{e}^{-\omega^2/2}. \tag{2.19}$$

(ii) Now consider the functions $g(t) = \mathrm{e}^{-t^2}$ and $h(t) = \mathrm{e}^{-(t^2+2t)/2}$. Both of these functions can be written in terms of $f(t)$ as

$$g(t) = f(t\sqrt{2}), \qquad h(t) = \mathrm{e}^{-((t+1)^2-1)/2} = \mathrm{e}^{1/2}f(t+1), \tag{2.20}$$

which the transformation rules of Fourier transforms imply

$$\hat{g}(\omega) = \sqrt{\pi}\mathrm{e}^{-\omega^2/4}, \quad \hat{h}(\omega) = \sqrt{2\pi}\mathrm{e}^{-\omega^2/2+\mathrm{i}\omega+1/2} \tag{2.21}$$

2.3 Convolution theorem

Given that the Fourier transform turns differentiation into multiplication by $\mathrm{i}\omega$, it is natural to ask which other operations it simplifies. Here we will examine its interaction with the *convolution* of two functions.

Definition 2.3. The *convolution* of two functions $f, g : \mathbb{R} \to \mathbb{C}$ is

$$(f * g)(t) \doteq \int_{-\infty}^{\infty} f(u)g(t-u)\,\mathrm{d}u. \tag{2.22}$$

Admittedly, the convolution is not immediately straightforward to understand, but if we break the definition down a bit we can get some intuition.

First of all, we can plot f as a function of u since there is no t-dependence. Then we can plot $g(-u)$ on top of this, which would correspond to $t = 0$ in the definition. The $t = 0$ convolution is then the area under the curve $f(u)g(-u)$, so the more $g(-u)$ and $f(u)$ overlap, the larger the convolution is. Finally, we can introduce the t-dependence by sliding the graph of $g(-u)$ across $f(u)$ to calculate the area under $f(u)g(t-u)$, calculating the overlap for different translations of $g(-u)$. The following example illustrates this in a simple case.

Example 2.3. Consider the function $f(t) = \chi_{[-1,1]}(t)$. The self-convolution $f * f$ is given by

$$(f * f)(t) = \int_{-\infty}^{\infty} \chi_{[-1,1]}(u)\chi_{[-1,1]}(t-u)\,\mathrm{d}u. \tag{2.23}$$

The first term in the integrand is non-zero when $-1 \leq u \leq 1$, while the second term is non-zero when $-1 \leq t-u \leq 1$, which we can rewrite as $-1+t \leq u \leq 1+t$. Since both of these need to be satisfied for the integrand to be non-zero, we must look at when these two ranges overlap. Firstly, if $|t| > 2$, then the two ranges do not overlap and $(f * f)(t) = 0$. Now take $-2 < t < 2$, then the overlap is

$$[-1,1] \cap [-1+t, 1+t] = [\max\{-1, -1+t\}, \min\{1, 1+t\}] \doteq [u_1, u_2]. \tag{2.24}$$

That is, if $t < 0$ then the intersection is $[-1, 1+t]$, while $t > 0$ gives $[-1+t, 1]$. Therefore, for $-2 < t < 2$ the convolution is

$$(f * f)(t) = \int_{u_1}^{u_2} \mathrm{d}u = \min\{1, 1+t\} - \max\{-1, -1+t\} = \begin{cases} 2-t, & t \geq 0 \\ 2+t & t < 0. \end{cases} \tag{2.25}$$

This is just $2 - |t|$, so we can write the convolution as

$$(f * f)(t) = (2 - |t|)\chi_{[-2,2]}(t), \tag{2.26}$$

where we've included $\chi_{[-2,2]}(t)$ to ensure it is zero outside of $[-2,2]$.

Now we turn to how the convolution behaves under Fourier transformation. First note that if $f, g \in L^1(\mathbb{R})$, we can show that $f * g$ also belongs to $L^1(\mathbb{R})$. Concretely, we have

$$\int_{-\infty}^{\infty} |(f*g)(t)|\,\mathrm{d}t = \int_{-\infty}^{\infty} \left| \int_{-\infty}^{\infty} f(u)g(t-u)\,\mathrm{d}u \right| \mathrm{d}t \leq \int_{\mathbb{R}^2} |f(u)||g(t-u)|\,\mathrm{d}u\,\mathrm{d}t. \tag{2.27}$$

We can eliminate t by changing variables to $s = t - u$, yielding

$$\int_{-\infty}^{\infty} |(f*g)(t)|\,\mathrm{d}t \leq \left(\int_{-\infty}^{\infty} |f(u)|\,\mathrm{d}u \right) \left(\int_{-\infty}^{\infty} |g(s)|\,\mathrm{d}s \right), \tag{2.28}$$

which is finite by assumption. Therefore, $f * g \in L^1(\mathbb{R})$ and we can attempt to Fourier transform it. In doing so, we arrive at

$$\begin{aligned} \widehat{(f*g)}(\omega) &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} f(u)g(t-u)\,\mathrm{d}u \right) \mathrm{e}^{-\mathrm{i}\omega t}\,\mathrm{d}t, \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(u)g(s)\mathrm{e}^{-\mathrm{i}\omega(s+u)}\,\mathrm{d}u\,\mathrm{d}s, \end{aligned} \tag{2.29}$$

by making the change of variables $s = t - u$. This integral factorises nicely as

$$\begin{aligned} \widehat{(f*g)}(\omega) &= \left(\int_{-\infty}^{\infty} f(u)\mathrm{e}^{-\mathrm{i}\omega u}\,\mathrm{d}u \right) \left(\int_{-\infty}^{\infty} g(s)\mathrm{e}^{-\mathrm{i}\omega s}\,\mathrm{d}s \right), \\ &= \hat{f}(\omega)\hat{g}(\omega). \end{aligned} \tag{2.30}$$

As such, convolution of functions gets mapped to products of Fourier transforms. For our purposes, this is the defining feature of the convolution: when we see a product of Fourier transforms, we should immediately think of the convolution.

Example 2.4. Using the previous example, we have

$$\mathcal{F}\left[(2-|t|)\chi_{[-2,2]}(t)\right] = \mathcal{F}[\chi_{[-1,1]}(t)]^2 = \frac{4\sin^2\omega}{\omega^2}. \tag{2.31}$$

2.4 Fourier inversion theorem

The Fourier transform is the natural analogue of the Fourier coefficients for periodic functions, but in that case we were also able to use these coefficients to reconstruct f via the Fourier series. We can also do this with the Fourier transform.

First we will sketch the idea and then state the theorem more concretely. Suppose we have an $L^1(\mathbb{R})$ function $f : \mathbb{R} \to \mathbb{C}$ that we wish to write down a Fourier series for. We can do this by defining f_T to be f restricted to the range $[-T/2, T/2]$ and extended periodically. Then, assuming the convergence of the Fourier series, we have

$$f_T(t) = \frac{1}{2\pi} \sum_{n\in\mathbb{Z}} \Omega \mathrm{e}^{\mathrm{i}n\Omega t} g(n\Omega), \qquad g(\omega) = \int_{-T/2}^{T/2} f(\tau)\mathrm{e}^{-\mathrm{i}\omega\tau}\,\mathrm{d}\tau, \tag{2.32}$$

where we've written the usual $1/T$ factor as $\Omega/2\pi$. Recall that the Riemann integral can be obtained by sampling a function at a number of equally distanced points, multiplying these samples by the gap between points and then taking the limit to close the gap. This is essentially what is going on here: the gap between points is given by Ω and we are sampling $e^{i\omega t}g(\omega)$ at $\omega = n\Omega$. The only caveat is that $g(\omega)$ changes as T increases due to the integration range being $[-T/2, T/2]$. We can get around this for T suitably large, by writing $g(\omega)$ as

$$g(\omega) = \hat{f}(\omega) - \int_{-\infty}^{-T/2} f(\tau)e^{-i\omega\tau}\,d\tau - \int_{T/2}^{\infty} f(\tau)e^{-i\omega\tau}\,d\tau, \tag{2.33}$$

and noting that these two corrections to $\hat{f}(\omega)$ must vanish as $T \to \infty$ since $f \in L^1(\mathbb{R})$. This leaves us with $g(\omega) \to \hat{f}(\omega)$ as $T \to \infty$ and we obtain

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \hat{f}(\omega)e^{i\omega t}\,d\omega. \tag{2.34}$$

This is our analogue of the Fourier series, only now it is an integral that sums over every possible frequency ω. In doing this, we've been quite sketchy with the details, but the rigorous result is as follows.

Theorem 2.1. Let $f : \mathbb{R} \to \mathbb{C}$ be a continuous absolutely integrable function such that $\hat{f}$ is also continuous and absolutely integrable. Then

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \hat{f}(\omega)e^{i\omega t}\,d\omega. \tag{2.35}$$

This is known as the *Fourier inversion theorem* and allows us to reconstruct $f(t)$ from its Fourier transform. There are various ways to prove this rigorously, but they are outside the scope of this course.

The fact that the inverse Fourier transform looks like the usual Fourier transform with $\omega \mapsto -\omega$ has an important consequence. Given a function's Fourier transform $\hat{f}(\omega)$, we can try to Fourier transform this as usual

$$\mathcal{F}[\hat{f}] = \int_{-\infty}^{\infty} \hat{f}(\omega)e^{-i\omega t}\,d\omega. \tag{2.36}$$

But as we've just remarked, this is just the inverse transform up to a minus sign in the exponent and a factor of 2π. We therefore get

$$\mathcal{F}^2[f] = \mathcal{F}[\hat{f}] = 2\pi f(-t). \tag{2.37}$$

In other words, if we recognise a function $g(t)$ as the Fourier transform of a function $f(t)$, then $\hat{g}(\omega) = 2\pi f(-\omega)$. This is sometimes referred to as the ***duality property*** of Fourier transforms.

2.5 Plancherel's theorem

Parseval's theorem also has a natural extension to Fourier transforms, only now it relates two integrals rather than an integral and a sum. To state this, we need to return to L^2 functions.

Definition 2.4. The space $L^2(\mathbb{R})$ consists of all functions $f : \mathbb{R} \to \mathbb{C}$ such that

$$\int_{-\infty}^{\infty} |f(t)|^2 \, \mathrm{d}t < \infty. \tag{2.38}$$

Like before, $L^2(\mathbb{R})$ is an inner product space with the inner product

$$\langle f, g \rangle = \int_{-\infty}^{\infty} f(t)\overline{g(t)} \, \mathrm{d}t, \tag{2.39}$$

satisfying the Cauchy-Schwarz inequality $|\langle f, g \rangle| \leq \sqrt{\langle f, f \rangle \langle g, g \rangle}$. Crucially, $L^2(\mathbb{R})$ is not a subspace of $L^1(\mathbb{R})$, as the following example illustrates.

Example 2.5. While many functions belong to both $L^1(\mathbb{R})$ and $L^2(\mathbb{R})$, we will construct two examples that only belong to one of the two spaces. Consider the two functions

$$f_1(x) = \begin{cases} \frac{1}{\sqrt{x}}, & 0 < x \leq 1, \\ 0, & \text{otherwise.} \end{cases} \qquad f_2(x) = \begin{cases} \frac{1}{x}, & x \geq 1, \\ 0, & \text{otherwise.} \end{cases} \tag{2.40}$$

The L^1-norm of f_1 is given by $\int_0^1 1/\sqrt{x} \, \mathrm{d}x = 2$, while the square integral is

$$\int_{-\infty}^{\infty} |f_1(x)|^2 \, \mathrm{d}x = \int_0^1 \frac{1}{x} \, \mathrm{d}x = [\log x]_0^1, \tag{2.41}$$

which is infinite. Therefore, f_1 belongs to $L^1(\mathbb{R})$, but not $L^2(\mathbb{R})$. For f_2 it is the opposite: it belongs to $L^2(\mathbb{R})$ but not $L^1(\mathbb{R})$.

Theorem 2.2. Let $f, g \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$. Then

$$\int_{-\infty}^{\infty} f(t)\overline{g(t)} \, \mathrm{d}t = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(\omega)\overline{\hat{g}(\omega)} \, \mathrm{d}\omega \tag{2.42}$$

and

$$\int_{-\infty}^{\infty} |f(t)|^2 \, \mathrm{d}t = \frac{1}{2\pi} \int_{-\infty}^{\infty} |\hat{f}(\omega)|^2 \, \mathrm{d}\omega. \tag{2.43}$$

Proof. We will assume that f and g are both continuous so the Fourier inversion theorem holds. In this case, we can write

$$\overline{g(t)} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \overline{\hat{g}(\omega)} \mathrm{e}^{-\mathrm{i}\omega t} \, \mathrm{d}\omega, \tag{2.44}$$

by moving the complex conjugation into the integrand. Substituting this into the left-hand side

of eq. (2.42) gives

$$\begin{aligned}\int_{-\infty}^{\infty} f(t)\overline{g(t)}\,\mathrm{d}t &= \int_{-\infty}^{\infty} f(t)\left(\frac{1}{2\pi}\int_{-\infty}^{\infty}\overline{\hat{g}(\omega)}\mathrm{e}^{-\mathrm{i}\omega t}\,\mathrm{d}\omega\right)\mathrm{d}t,\\ &= \int_{-\infty}^{\infty}\left(\frac{1}{2\pi}\int_{-\infty}^{\infty} f(t)\mathrm{e}^{-\mathrm{i}\omega t}\,\mathrm{d}t\right)\overline{\hat{g}(\omega)}\,\mathrm{d}\omega.\end{aligned} \tag{2.45}$$

Here we're allowed to exchange the order of integration since both $f(t)$ and $\hat{g}(\omega)$ are assumed to be in $L^1(\mathbb{R})$. We now recognise the integral over t as the Fourier transform of f, therefore

$$\int_{-\infty}^{\infty} f(t)\overline{g(t)}\,\mathrm{d}t = \frac{1}{2\pi}\int_{-\infty}^{\infty}\hat{f}(\omega)\overline{\hat{g}(\omega)}\,\mathrm{d}\omega. \tag{2.46}$$

Taking $g = f$ then gives the second result. □

Plancherel's theorem states that the Fourier transform preserves the $L^2(\mathbb{R})$ inner product up to a factor of 2π. Consequently, the Fourier transform of an $L^2(\mathbb{R})$ function also belongs to $L^2(\mathbb{R})$. Contrast this with the $L^1(\mathbb{R})$ case where the Fourier transform is often not absolutely integrable, with the Fourier transform of $\chi_{[-a,a]}(t)$ being a simple example.

Example 2.6. (i) We saw that $\chi_{[-a,a]}(t)$ has Fourier transform $2\sin a\omega/\omega$ earlier. Taking $f(t) = \chi_{[-a,a]}(t)$ and $g(t) = \chi_{[-b,b]}(t)$, Plancherel's formula states

$$\int_{-\infty}^{\infty}\chi_{[-a,a]}(t)\chi_{[-b,b]}(t)\,\mathrm{d}t = \frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{4\sin a\omega\sin b\omega}{\omega^2}\,\mathrm{d}\omega. \tag{2.47}$$

The right-hand side is difficult to compute, but the left-hand side is easy. The integrand is non-zero only when $|t| < \min\{a,b\}$, so

$$\int_{-\infty}^{\infty}\frac{\sin a\omega\sin b\omega}{\omega^2}\,\mathrm{d}\omega = \pi\min\{a,b\}. \tag{2.48}$$

(ii) The function $f(t) = \mathrm{e}^{-|t|}$ has Fourier transform $\hat{f}(\omega) = 2/(1+\omega^2)$. Integrating $1/(1+\omega^2)^2$ over $\mathbb{R}$ is doable via substitution or integration by parts, but using Plancherel's theorem it is even easier. Namely, we have

$$\int_{-\infty}^{\infty}\frac{\mathrm{d}\omega}{(1+\omega^2)^2} = \frac{\pi}{2}\int_{-\infty}^{\infty}\mathrm{e}^{-2|t|}\,\mathrm{d}t = \pi\int_0^{\infty}\mathrm{e}^{-2t}\,\mathrm{d}t = \frac{\pi}{2}. \tag{2.49}$$

If you're brave, you can do the integral directly to confirm this matches.

3 The Laplace transform

3.1 Motivation and definition

While the Fourier transform is useful for solving myriad problems, it is limited in some aspects. One of the more important limitations is the fact that we cannot calculate the Fourier transform of many functions that grow as $t \to \infty$. This makes it unreliable when we suspect solutions that grow exponentially, for example. However, there is an easy way around this. We can simply modify the definition of the Fourier transform in such a way to dampen exponential functions as $t \to \infty$, thus giving us well defined transforms. This leads us to the following definition.

Definition 3.1. Let $f : \mathbb{R}_{\geq 0} \to \mathbb{C}$ be a function that grows at most exponentially as $t \to \infty$. The *Laplace transform* of f is given by

$$F(s) = \mathcal{L}[f] = \int_0^\infty f(t)\mathrm{e}^{-st}\,\mathrm{d}t. \tag{3.1}$$

Before diving into some examples, some comments are in order.

- Since the range of integration is $[0, \infty)$, we only require f to be defined for $t \geq 0$, as opposed to the Fourier transform. This makes the Laplace transform particularly well suited to dealing with *initial value problems*, but ill suited to problems involving functions defined over all of $\mathbb{R}$.
- The variable s is generally taken to be complex, but we need to be careful about for which values of s the integral exists. Below we will make this explicit in several examples.
- The phrase 'grows at most exponentially' means that there exist $M, N \in \mathbb{R}$ such that for all t, we have

  $$|f(t)| \leq M\mathrm{e}^{Nt}. \tag{3.2}$$

  This will guarantee that the Laplace transform exists for $\mathrm{Re}(s) > N$. We will also call these functions of *exponential-type*.

Since we only care about the behaviour of a function $f(t)$ for positive t, it is useful to define the *Heaviside step function* $H(t)$ to be

$$H(t) = \begin{cases} 1, & t \geq 0, \\ 0, & t < 0. \end{cases} \tag{3.3}$$

Herein, whenever we reference a function $f(t)$ defined over all of $\mathbb{R}$, we really are talking about $f(t)H(t)$. Similarly, we can think of a function $f : \mathbb{R}_{\geq 0} \to \mathbb{C}$ as a function defined over all of $\mathbb{R}$ that vanishes for $t < 0$.

Example 3.1. Here we will list several useful Laplace transforms, noting any restrictions on s. It is a good exercise to check these yourself.

(i) $\mathcal{L}[e^{\alpha t}] = \int_0^\infty e^{(\alpha - s)t}\,dt = \dfrac{1}{s-\alpha}, \qquad \mathrm{Re}(s) > \mathrm{Re}(\alpha).$

(ii) $\mathcal{L}[\cos t] = \int_0^\infty e^{-st}\cos t\,dt = \dfrac{1}{2}\int_0^\infty e^{(i-s)t} + e^{-(i+s)t}\,dt = \dfrac{s}{s^2+1}, \quad \mathrm{Re}(s) > 0.$

(iii) $\mathcal{L}[\sin t] = \int_0^\infty e^{-st}\sin t\,dt = \dfrac{1}{2i}\int_0^\infty e^{(i-s)t} - e^{-(i+s)t}\,dt = \dfrac{1}{s^2+1}, \quad \mathrm{Re}(s) > 0.$

(iv) $\mathcal{L}[H(t)] = \int_0^\infty e^{-st}\,dt = \dfrac{1}{s}, \quad \mathrm{Re}(s) > 0.$

(v) $\mathcal{L}[t] = \int_0^\infty te^{-st}\,dt = \dfrac{1}{s^2}, \quad \mathrm{Re}(s) > 0.$

(vi) $\mathcal{L}[t^n] = \int_0^\infty t^n e^{-st}\,dt = (-1)^n \dfrac{d^n}{ds^n}\left[\int_0^\infty e^{-st}\,dt\right] = \dfrac{n!}{s^{n+1}}, \quad \mathrm{Re}(s) > 0.$

Note that n is a positive integer in the final example. For non-integer values of n, we can make the substitution $q = st$ to give

$$\mathcal{L}[t^n] = \int_0^\infty \left(\frac{q^n}{s^n}\right) e^{-q}\frac{dq}{s} = \frac{1}{s^{n+1}}\int_0^\infty q^n e^{-q}\,dq \doteq \frac{\Gamma(n+1)}{s^{n+1}}, \tag{3.4}$$

where $\Gamma(n+1)$ is a generalisation of the factorial called the *Gamma function*.

Similar to how the Fourier transforms of $L^1(\mathbb{R})$ functions vanish as $\omega \to \infty$, the Laplace transforms of exponential type functions also decay appropriately. Concretely, we have the following result.

Theorem 3.1. The Laplace transform F of an exponential-type function f obeys

$$\lim_{s\to\infty} F(s) = 0. \tag{3.5}$$

Proof. Since f is of exponential-type, we have $|f(t)| \le Me^{Nt}$ for all t. Hence

$$|F(s)| \le \int_0^\infty |f(t)||e^{-st}|\,dt = M\int_0^\infty e^{Nt}e^{-st}\,dt, \tag{3.6}$$

where we're taking s to be real in order to take the limit. Computing this integral we see that

$$\lim_{s\to\infty}|F(s)| = \lim_{s\to\infty}\left(\frac{M}{s-N}\right) = 0. \tag{3.7}$$

□

3.2 Transformation properties

Just like the Fourier transform, we can use substitution of variables to understand the transformation rules for Laplace transform. These derivations are much the same, up to the caveat that changing the integration range to $[0,\infty)$ introduces a few technicalities we need to deal with.

First of all, the Laplace transforms of $\alpha f(t) + \beta g(t)$ and $f(at)$ are completely analogous to the Fourier transform case. The shifted case $f(t-a)$ with $a > 0$ is similar, but we have to take into account that the Laplace transform only knows about the values of $f(t)$ for $t > 0$. We should therefore multiply $f(t-a)$ by the step function $H(t-a)$ to ensure no values of $f(t)$ for $t < 0$ contribute to the integral. Doing so gives

$$\mathcal{L}[f(t-a)H(t-a)] = \int_0^\infty f(t-a)H(t-a)\mathrm{e}^{-st}\,\mathrm{d}t = \int_a^\infty f(t-a)\mathrm{e}^{-st}\,\mathrm{d}t, \tag{3.8}$$

which changing variables to $\tau = t - a$ shows

$$\mathcal{L}[f(t-a)H(t-a)] = \mathrm{e}^{-as}\int_0^\infty f(\tau)\mathrm{e}^{-s\tau}\,\mathrm{d}\tau = \mathrm{e}^{-as}F(s). \tag{3.9}$$

In other words, delaying a function by a time $a > 0$ changes the Laplace transform by a factor of e^{-as}.

Now consider the Laplace transform of the derivative $f'(t)$. We can integrate by parts to obtain

$$\mathcal{L}[f'(t)] = \int_0^\infty f'(t)\mathrm{e}^{-st}\,\mathrm{d}t = \Big[f(t)\mathrm{e}^{-st}\Big]_0^\infty + s\int_0^\infty f(t)\mathrm{e}^{-st}\,\mathrm{d}t. \tag{3.10}$$

By hypothesis, $f(t)$ grows at most exponentially, so for large enough s we know that $f(t)\mathrm{e}^{-st}$ vanishes at infinity, but we still pick up the lower limit. Hence

$$\mathcal{L}[f'(t)] = sF(s) - f(0), \tag{3.11}$$

where $f(0)$ is understood to mean $f(0^-)$ when 0 is a point of discontinuity. Assuming $f(t)$ is n-times differentiable, we can extend this

$$\begin{aligned}\mathcal{L}[f^{(n)}(t)] &= \int_0^\infty f^{(n)}(t)\mathrm{e}^{-st}\,\mathrm{d}t = \Big[f^{(n-1)}(t)\mathrm{e}^{-st}\Big]_0^\infty + s\mathcal{L}[f^{(n-1)}(t)],\\ &= -f^{(n-1)}(0) + s\Big(\Big[f^{(n-2)}(t)\mathrm{e}^{-st}\Big]_0^\infty + s\mathcal{L}[f^{(n-2)}(t)]\Big).\end{aligned} \tag{3.12}$$

By repeatedly integrating by parts we keep lowering the derivative of f and introduce a factor of s. Keeping track of all the boundary terms, we see

$$\begin{aligned}\mathcal{L}[f^{(n)}(t)] &= s^nF(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \ldots - sf^{(n-2)}(0) - f^{(n-1)}(0),\\ &= s^nF(s) - \sum_{k=1}^{n} s^{n-k}f^{(k-1)}(0).\end{aligned} \tag{3.13}$$

Up to the boundary terms, we see the Laplace transform also turns differentiation into multiplication. In the opposite direction, we have

$$\mathcal{L}[t^nf(t)] = (-1)^n\frac{\mathrm{d}^n}{\mathrm{d}s^n}\left(\int_0^\infty f(t)\mathrm{e}^{-st}\,\mathrm{d}t\right) = (-1)^n\frac{\mathrm{d}^nF}{\mathrm{d}s^n}. \tag{3.14}$$

function	Laplace transform
$f(t)$	$F(s)$
$\alpha f(t)+\beta g(t)$	$\alpha F(s)+\beta G(s)$
$f(at)$	$\frac{1}{a}F\left(\frac{s}{a}\right)$
$\mathrm{e}^{-bt}f(t)$	$F(s+b)$
$f(t-a)H(t-a)$	$\mathrm{e}^{-as}F(s)$
$t^n f(t)$	$(-1)^n F^{(n)}(s)$
$f'(t)$	$sF(s)-f(0)$
$f^{(n)}(t)$	$s^n F(s)-\sum_{k=1}^{n} s^{n-k} f^{(k-1)}(0)$
$\int_0^t f(u)\,\mathrm{d}u$	$s^{-1}F(s)$
$(f*g)(t)$	$F(s)G(s)$

Table 2: The transformation properties of Laplace transforms. Here $a>0$, $b\in\mathbb{R}$ and α,β are arbitrary complex numbers

Example 3.2. Let $f(t)=\sin t/t$. By eq. (3.14), the Laplace transform is related to the Laplace transform of $\sin t$ by

$$\mathcal{L}[\sin t]=\frac{1}{s^2+1}=-\frac{\mathrm{d}F}{\mathrm{d}s}. \tag{3.15}$$

Therefore, $F(s)$ is given by the integral

$$F(s)=-\int\frac{\mathrm{d}s}{s^2+1}=-\tan^{-1}(s)+C. \tag{3.16}$$

We can eliminate the constant of integration by noting that $F(s)\to 0$ as $s\to\infty$. Combining this with $\tan^{-1}(s)\to\pi/2$, we see that $C=\pi/2$ and the Laplace transform is

$$\mathcal{L}\left[\frac{\sin t}{t}\right]=\frac{\pi}{2}-\tan^{-1}(s) \tag{3.17}$$

The convolution theorem also holds for Laplace transform, but since we are only interested in the value of functions on $[0,\infty)$, we have a nice simplification. Suppose $f(t)$ and $g(t)$ are two functions that are zero for $t<0$. Then the convolution takes the form

$$(f*g)(t)=\int_{-\infty}^{\infty}f(u)g(t-u)\,\mathrm{d}u=\int_0^t f(u)g(t-u)\,\mathrm{d}u, \tag{3.18}$$

since $f(u)g(t-u)$ is only non-zero when both $u\geq 0$ and $t-u\geq 0$.

Theorem 3.2. If $f,g:\mathbb{R}_{\geq 0}\to\mathbb{C}$ are functions that grow at most exponentially, then

$$\mathcal{L}[(f*g)(t)]=\mathcal{L}\left[\int_0^t f(u)g(t-u)\,\mathrm{d}u\right]=F(s)G(s). \tag{3.19}$$

Proof. We proceed similarly to the Fourier transform case. First note that if both $f(t)$ and $g(t)$ grow at most exponentially, then so does $(f*g)(t)$. The integral expression for the Laplace

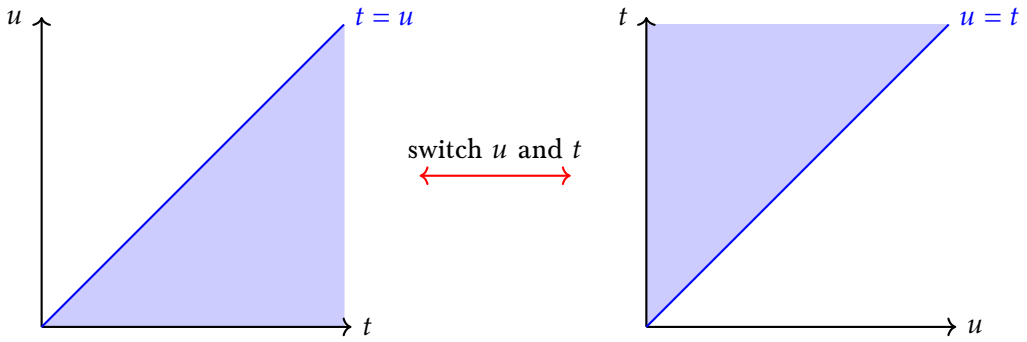


Figure 6: *Left:* The shaded area represents the domain of integration in eq. (3.20). *Right:* Changing the order of integration flips the domain so the limits change to $u \leq t < \infty$ and $0 \leq u < \infty$.

transform of $(f * g)(t)$ therefore holds. Explicitly, we have

$$\mathcal{L}[(f*g)(t)] = \int_0^\infty \left(\int_0^t f(u) g(t-u)\, \mathrm{d}u \right) \mathrm{e}^{-st}\, \mathrm{d}t. \tag{3.20}$$

We can switch the order of integration, but we have to be careful about the range of the integrals. By changing the order so we do the t integral first, we must reparametrise the domain of integration as $\{(t,u) : u \leq t < \infty, 0 \leq u < \infty\}$ (see fig. 6). Therefore, the Laplace transform is

$$\begin{aligned} \mathcal{L}[(f*g)(t)] &= \int_0^\infty \int_u^\infty f(u) g(t-u) \mathrm{e}^{-st}\, \mathrm{d}t\, \mathrm{d}u, \\ &= \int_0^\infty \int_0^\infty f(u) g(\tau) \mathrm{e}^{-s(\tau+u)}\, \mathrm{d}\tau\, \mathrm{d}u, \end{aligned} \tag{3.21}$$

where we've changed the t variable to $\tau = t - u$ in the final line. This integral factorises as

$$\mathcal{L}[(f*g)(t)] = \left(\int_0^\infty f(u) \mathrm{e}^{-su}\, \mathrm{d}u \right) \left(\int_0^\infty g(\tau) \mathrm{e}^{-s\tau}\, \mathrm{d}\tau \right) = F(s)G(s), \tag{3.22}$$

giving the result. □

Example 3.3. (i) The convolution of any function $f(t)$ with the step function $H(t)$ is simply

$$(f*H)(t) = \int_0^t f(u)\, \mathrm{d}u. \tag{3.23}$$

From the previous examples, we know that $\mathcal{L}[H(t)] = 1/s$, so the convolution theorem tells us

$$\mathcal{L}\left[\int_0^t f(u)\, \mathrm{d}u \right] = \frac{F(s)}{s}. \tag{3.24}$$

(ii) Consider the functions $f(t) = t$ and $g(t) = \mathrm{e}^t$ for $t \geq 0$. The convolution of these two

functions is given by

$$(f * g)(t) = \int_0^t u\mathrm{e}^{t-u}\,\mathrm{d}u = \mathrm{e}^t \int_0^t u\mathrm{e}^{-u}\,\mathrm{d}u = \mathrm{e}^t(-t\mathrm{e}^{-t} + 1 - \mathrm{e}^{-t}) = \mathrm{e}^t - t - 1. \tag{3.25}$$

From the previous example, we already know how to compute the Laplace transform of the right-hand side. It is given by

$$\mathcal{L}[\mathrm{e}^t - t - 1] = \frac{1}{s-1} - \frac{1}{s^2} - \frac{1}{s} = \frac{1}{(s-1)s^2} = \mathcal{L}[t]\mathcal{L}[\mathrm{e}^t], \tag{3.26}$$

matching our expectations from the convolution theorem.

Before we move on, we should note that the duality property of Fourier transforms does not generalise nicely to Laplace transforms. In particular, we have

$$\mathcal{L}^2[f(t)](u) = \int_0^\infty \int_0^\infty f(t)\mathrm{e}^{-st-us}\,\mathrm{d}t\,\mathrm{d}s = \int_0^\infty \frac{f(t)}{t+u}\,\mathrm{d}t, \tag{3.27}$$

by assuming we can change the order of integration and integrating over s. Therefore, we cannot simply read the formula sheet for Laplace transforms 'backwards' to find the Laplace transform of $F(t) = \mathcal{L}[f]$. Instead, we must (sadly) calculate it directly.

3.3 Solving ordinary differential equations

One of the main applications of the Laplace transform is using it to solve ordinary differential equations. While the Fourier transform can often be used for this purpose, the Laplace transform has some advantages. The first, which we already mentioned, is that the Laplace transform can deal with exponentially growing solutions, whereas the Fourier transform cannot. Secondly, although the boundary terms in $\mathcal{L}[y^{(n)}]$ may look cumbersome, they allow us to incorporate the initial conditions of a problem easily. This gives us better control over the solution when compared with the solution obtained from using the Fourier transform.

Example 3.4. Consider the second order ODE

$$y''(t) + y(t) = t\mathrm{e}^t, \tag{3.28}$$

subject to $y(0) = 1$ and $y'(0) = 0$. With these initial conditions, the Laplace transform of $y''(t)$ is given by $s^2Y(s) - s$, so taking the Laplace transform of the ODE gives

$$(s^2Y(s) - s) + Y(s) = \mathcal{L}[t\mathrm{e}^t] = -\frac{\mathrm{d}}{\mathrm{d}s}\left(\frac{1}{s-1}\right) = \frac{1}{(s-1)^2}, \tag{3.29}$$

where we've used the fact $\mathcal{L}[tf(t)] = -F'(s)$ to obtain $\mathcal{L}[t\mathrm{e}^t]$. We now collect all terms involving $Y(s)$ on one side of the equation as

$$(s^2+1)Y(s) = \frac{1}{(s-1)^2} + s, \tag{3.30}$$

which then gives

$$Y(s) = \frac{1}{(s-1)^2(s^2+1)} + \frac{s}{s^2+1}. \tag{3.31}$$

All that is left is to invert this to find $y(t)$. However, while the second term is just the Laplace transform of $\cos t$, we need to do some more work to invert the first.

In order to invert $Y(s)$ in the above example, we want to rewrite it in terms of Laplace transforms we already know. Crucially, $Y(s)$ is a *rational function*, which makes this procedure fairly easy. A rational function is a function that can be written as $R(s) = P(s)/Q(s)$ where P and Q are polynomials. Assuming that $\deg(P) < \deg(Q)$, we can always perform partial fraction reduction to break up $R(s)$ into a sum of terms like $A/(s-\alpha)^n$. Let's do a simple example to recall how to do partial fractions.

Example 3.5. Consider the rational function $R(s) = s/(s-1)(s+1)^2$. First, we write out the expansion we wish to find

$$\frac{s}{(s-1)(s+1)^2} = \frac{A}{s-1} + \frac{B}{(s+1)} + \frac{C}{(s+1)^2}. \tag{3.32}$$

Now we can multiply the whole expression by $(s-1)(s+1)^2$ to eliminate the fractions

$$s = A(s+1)^2 + B(s-1)(s+1) + C(s-1). \tag{3.33}$$

There are several ways of finding A, B and C. We could expand the right-hand side and then compare coefficients to determine the unknowns. Alternatively, we can take clever values of s to obtain simpler equations and solve those instead. For example, if we take $s = 1$, all terms except for the first one vanish, so

$$(s=1): \quad 1 = 4A \Longrightarrow A = \frac{1}{4}. \tag{3.34}$$

Similarly, taking $s = -1$ makes all terms except the final one vanish, leading to

$$(s=-1): \quad -1 = -2C \Longrightarrow C = \frac{1}{2}. \tag{3.35}$$

Unfortunately, no choice of s will cause both the first and third term to vanish, so we can just take $s = 0$ to get

$$(s=0): \quad 0 = \frac{1}{4} - B - \frac{1}{2} \Longrightarrow B = -\frac{1}{4}. \tag{3.36}$$

Therefore, $R(s)$ can be written as

$$R(s) = \frac{s}{(s-1)(s+1)^2} = \frac{1}{4(s-1)} - \frac{1}{4(s+1)} + \frac{1}{2(s+1)^2}. \tag{3.37}$$

In this example, we factorised $Q(s)$ into linear factors, but this isn't actually necessary. For

example, consider the function $R(s) = 1/(s+1)(s^2+2s+2)$. The factor s^2+2s+2 has two complex roots, so instead of factorising it into linear factors we can keep it as is. Then the partial fraction decomposition takes the form

$$\frac{1}{(s+1)(s^2+2s+2)} = \frac{A}{s+1} + \frac{Bs+C}{s^2+2s+2}. \tag{3.38}$$

The price we pay in keeping s^2+2s+2 as a quadratic is that we need to make sure to include the term Bs in the expansion. We can then proceed as usual to get

$$\frac{1}{(s+1)(s^2+2s+2)} = \frac{1}{s+1} - \frac{s+1}{s^2+2s+2}. \tag{3.39}$$

For the rest of this chapter, we will assume that $Q(s)$ can always be factorised into linear factors with *real* roots and quadratic factors with *complex* roots. With this assumption, inverting the Laplace transform is reasonably straightforward. For the linear factors, we can use

$$\mathcal{L}^{-1}\left[\frac{n!}{(s-a)^{n+1}}\right] = t^n \mathrm{e}^{at} \tag{3.40}$$

For the quadratic factors, we can always complete the square and use the inverse transforms

$$\mathcal{L}^{-1}\left[\frac{b}{(s+a)^2+b^2}\right] = \mathrm{e}^{-at}\sin bt, \quad \mathcal{L}^{-1}\left[\frac{s+a}{(s+a)^2+b^2}\right] = \mathrm{e}^{-at}\cos bt, \tag{3.41}$$

with $b \neq 0$. Both of these follow from the transformation rules we derived earlier, along with the results of example 3.1.

Example 3.4. *(continued).* Returning to the differential equation, we are now ready to invert $Y(s)$ to find the solution. We start by performing partial fractions as

$$\frac{1}{(s-1)^2(s^2+1)} = \frac{A}{s-1} + \frac{B}{(s-1)^2} + \frac{Cs+D}{s^2+1}, \tag{3.42}$$

which multiplying by $(s-1)^2(s^2+1)$ gives

$$1 = A(s-1)(s^2+1) + B(s^2+1) + (Cs+D)(s-1)^2. \tag{3.43}$$

Taking $s = 1$ causes the terms with A, C and D to vanish, therefore

$$(s=1): \quad 1 = 2B \Longrightarrow B = \frac{1}{2}. \tag{3.44}$$

Since s^2+1 vanishes at $s = \mathrm{i}$, we can substitute this in to get

$$(s=\mathrm{i}): \quad 1 = (C\mathrm{i}+D)(\mathrm{i}-1)^2 = (C\mathrm{i}+D)(-2\mathrm{i}) = 2C - 2\mathrm{i}D, \tag{3.45}$$

while $s = -\mathrm{i}$ leads to

$$(s=-\mathrm{i}): \quad 1 = (-C\mathrm{i}+D)(2\mathrm{i}) = 2C + 2\mathrm{i}D. \tag{3.46}$$

Adding these two equations gives $C = 1/2$ which in turn implies $D = 0$. Finally, taking $s = 0$ then gives $A = -1/2$, hence

$$\frac{1}{(s-1)^2(s^2+1)} = -\frac{1}{2(s-1)} + \frac{1}{2(s-1)^2} + \frac{s}{2(s^2+1)}. \tag{3.47}$$

Recalling that $Y(s)$ also included an additional $s/(s^2+1)$, we get the solution $y(t)$

$$y(t) = -\frac{1}{2}e^t + \frac{1}{2}te^t + \frac{3}{2}\cos t. \tag{3.48}$$

As a sanity check, we can easily see that $y(0) = 1$ and $y'(0) = 0$.

3.4 Initial and final value theorems

Suppose that we are given a Laplace transform of a function $f(t)$ that we don't know. While it would be good to be able to find the function, it is sometimes infeasible or difficult. Despite this, it turns out that we can understand the initial value of $f(t)$ and the value of $f(t)$ at large time scales purely from the Laplace transform. This is summarised by the following two theorems.

Theorem 3.3. If $f : \mathbb{R}_{\geq 0} \to \mathbb{C}$ has Laplace transform F, then

$$\lim_{t\to 0^+} f(t) = \lim_{s\to\infty} sF(s) \tag{3.49}$$

and

$$\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s), \tag{3.50}$$

provided the limits of $f(t)$ exist. These are the *initial value* and *final value* theorems respectively.

Proof. We start with the right-hand side and will show it equals the left. Using the integral expression for $F(s)$, we have

$$\lim_{s\to\infty} sF(s) = \lim_{s\to\infty} \int_0^\infty sf(t)e^{-st}\,dt. \tag{3.51}$$

Introducing the variable $u = st$, so $dt = du/s$, we get

$$\lim_{s\to\infty} sF(s) = \lim_{s\to\infty} \int_0^\infty f\left(\frac{u}{s}\right)e^{-u}\,du = f(0^+)\lim_{s\to\infty}\int_0^\infty e^{-u}\,du. \tag{3.52}$$

Here we've noted that $f(u/s) \to f(0^+)$ as $s \to \infty$ and pulled it out of the integral as a constant.[1] Computing the integral shows that

$$\lim_{s\to\infty} sF(s) = f(0^+) = \lim_{t\to 0^+} f(t), \tag{3.53}$$

giving the initial value theorem. A similar argument gives the final value theorem. □

[1] This step needs a little more justification in order to make it rigorous, but we will brush it under the rug.

The condition that the limit of $f(t)$ as $t \to \infty$ exists puts fairly strict conditions on the poles of $sF(s)$. These are the values of s for which $sF(s)$ is infinite. In particular, for the limit to exist, all poles of $sF(s)$ must belong to the half-plane $\mathbb{H} = \{s : \mathrm{Re}(s) < 0\}$. If this isn't the case, the limit of $sF(s)$ as $s \to 0$ may well be finite, while the limit of $f(t)$ as $t \to \infty$ won't exist.

Example 3.6. (i) Consider the function $f(t) = \sin t/t$. The limit as $t \to 0^+$ is given by

$$\lim_{t\to 0^+} \frac{\sin t}{t} = \lim_{t\to 0^+} \frac{\cos t}{1} = 1, \tag{3.54}$$

by l'Hôpital's rule. We've seen that the Laplace transform of $f(t)$ is given by $\pi/2 - \tan^{-1}(s)$, so the initial value theorem therefore states that

$$1 = \lim_{s\to\infty} s\left(\frac{\pi}{2} - \tan^{-1}(s)\right). \tag{3.55}$$

(ii) Suppose we are given the Laplace transform

$$F(s) = \frac{\pi}{s(s+\mathrm{e})(s+\sqrt{2})}. \tag{3.56}$$

Instead of trying to invert this, we can use the initial and final value theorems to learn about $f(t)$. The initial value theorem states

$$\lim_{t\to 0^+} f(t) = \lim_{s\to\infty} \frac{\pi}{(s+\mathrm{e})(s+\sqrt{2})} = 0, \tag{3.57}$$

while the final value theorem gives

$$\lim_{t\to\infty} f(t) = \lim_{s\to 0} \frac{\pi}{(s+\mathrm{e})(s+\sqrt{2})} = \frac{\pi}{\mathrm{e}\sqrt{2}}. \tag{3.58}$$

Since the only poles of $sF(s)$ are at $s = -\mathrm{e}$ and $s = -\sqrt{2}$, this will give the correct limit. As an exercise, you can invert $F(s)$ and check that the limits match.

3.5 The Dirac delta function

When modelling physical or electrical systems, we regularly need to model how the system reacts to a momentary impulse. For example, if we want to understand the motion of snooker balls colliding, we need to be able to model the moment of impact between the balls to accurately measure the change in momentum. In order to do this, it is useful to define the following tool.

Definition 3.2. The *Dirac delta function* δ (or *unit impulse*) is a generalised function characterised by the properties

$$\delta(t) = 0, \quad \forall t \neq 0 \tag{3.59}$$

and

$$\int_{-\infty}^{\infty} f(t)\delta(t)\,\mathrm{d}t = f(0), \tag{3.60}$$

for all functions $f : \mathbb{R} \to \mathbb{C}$.

As far as definitions go, this one is weird. How can a function be zero almost everywhere but somehow have an integral equal to 1? Simply put, it can't— the Dirac delta function isn't really a function and the definition above is heuristic. Mathematically, it is a ***generalised function*** or ***distribution***, both concepts that allow behaviour beyond regular functions. We won't dwell on either of these concepts, but we will show how the delta function can arise as a limit of familiar functions.

Consider the family of functions

$$D_a(t) = \frac{\chi_{[-a,a]}(t)}{2a}, \tag{3.61}$$

which has the property $\int_{-\infty}^{\infty} D_a(t)\,\mathrm{d}t = 1$ for any $a > 0$. As we take $a \to 0$, the limit of $D_a(t)$ for $t \neq 0$ is 0, while $D_a(0)$ diverges. This is precisely the behaviour we want, but we need to check this gives the correct property when integrated against a function f. For the sake of simplicity, let $f : \mathbb{R} \to \mathbb{R}$ be a continuous real-valued function, then we have

$$\min_{t\in[-a,a]} f(t) \leq \frac{1}{2a}\int_{-\infty}^{\infty} f(t)\chi_{[-a,a]}(t)\,\mathrm{d}t \leq \max_{t\in[-a,a]} f(t). \tag{3.62}$$

Since we know that f is continuous, we know that the minimum and maximum of $f(t)$ on the range $[-a, a]$ coincide under the limit $a \to 0$, meaning that both the upper and lower bound for the integral are equal when $a \to 0$. Therefore, we come to the conclusion that

$$\lim_{a\to 0}\left(\int_{-\infty}^{\infty} f(t)\frac{\chi_{[-a,a]}(t)}{2a}\,\mathrm{d}t\right) = f(0). \tag{3.63}$$

As such, we can say that the limit of $\frac{1}{2a}\chi_{[-a,a]}(t)$ is $\delta(t)$, as long as we're integrating against a function that is continuous at $t = 0$. Presenting $\delta(t)$ as a limit in this way is not unique, we have that

$$f_n(t) = \sqrt{\frac{n}{\pi}}\mathrm{e}^{-nt^2}, \quad g_n(t) = \frac{\sin nt}{\pi t}, \quad h_n(t) = \begin{cases} n(1-n|t|) & |t| < 1/n, \\ 0 & |t| \geq 1/n, \end{cases} \tag{3.64}$$

all tend to $\delta(t)$ as $n \to \infty$ in the same sense (i.e., when integrated against a function continuous at $t = 0$).

We can use the defining properties of the delta function to derive a few useful identities. First of all, we know that the integral over $\mathbb{R}$ of a delta function is equal to 1, but we also know that away from $t = 0$ the delta function is zero. Therefore, it follows that integrating over a range (a, b) will evaluate to 0 if $t = 0$ does not belong to $[a, b]$. In summary, we have

$$\int_a^b \delta(t)\,\mathrm{d}t = \begin{cases} 1, & 0 \in (a,b), \\ 0, & \text{otherwise.} \end{cases} \tag{3.65}$$

If we take the integration range to be $(-\infty, \tau]$, this implies

$$\int_{-\infty}^{\tau} \delta(t)\,\mathrm{d}t = H(\tau), \tag{3.66}$$

where $H(\tau)$ is the Heaviside step function. However, note that the integral in eq. (3.65) is ambiguous if one of the endpoints is equal to 0. Depending on the limiting function you use as a presentation for δ, you can argue for various values of the integral. To avoid this, we write

$$\int_{0^-}^{\infty} \delta(t)\,\mathrm{d}t = \lim_{h\to 0} \int_{-|h|}^{\infty} \delta(t)\,\mathrm{d}t = 1, \tag{3.67}$$

to indicate that $t = 0$ should be integrated over from below. Using this convention, the Laplace transform of $\delta(t)$ is given by

$$\mathcal{L}[\delta(t)] = \int_{0^-}^{\infty} \delta(t)\mathrm{e}^{-st}\,\mathrm{d}t = 1. \tag{3.68}$$

This is consistent with eq. (3.66). As we saw earlier, the integral of a function has Laplace transform

$$\mathcal{L}[H(t)] = \frac{1}{s}\mathcal{L}[\delta(t)] = \frac{1}{s}, \tag{3.69}$$

which agrees with the Laplace transform we derived for $H(t)$ earlier.

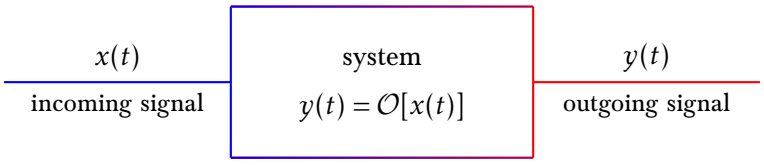
Importantly, the delta function acts as a unit for convolution. That is, we have $(f * \delta) = (\delta * f) = f$ for any (suitably well-behaved) function. We can verify this directly by noting

$$(f * \delta)(t) = \int_{-\infty}^{\infty} f(u)\delta(t-u)\,\mathrm{d}u = \int_{-\infty}^{\infty} f(t-\tau)\delta(\tau)\,\mathrm{d}\tau = f(t), \tag{3.70}$$

where we've used the substitution $\tau = t - u$. If we take $f(t)$ to be zero for $t < 0$, the same identity holds using the convention of making the lower limit 0^- we mentioned earlier.

3.6 Linear time-invariant systems

Quite often in engineering, we wish to analyse something made up of many interacting elements like a circuit. In these cases we supply an incoming signal $x(t)$ and wish to understand what the outgoing signal $y(t)$ is without understanding each interacting element separately. We call this a *system* and we think of it as an operator $\mathcal{O}$ which maps $x(t)$ to $y(t)$ and visualise it diagrammatically as below.



Right now, we're being very general: the operation $\mathcal{O}$ could send x to $x'(t)^2$, $\pi^{x(t)}$ or 0 for all we know. We therefore want to put some reasonable restrictions on the system so we can better analyse it.

Definition 3.3. A system $\mathcal{O} : x(t) \mapsto y(t)$ is *linear time-invariant (LTI)* if it satisfies the following properties.

- *Linearity.* For any two incoming signals $x_1(t)$ and $x_2(t)$ and any choice of constants $\alpha, \beta \in \mathbb{C}$, $\mathcal{O}$ satisfies

$$\mathcal{O}[\alpha x_1(t) + \beta x_2(t)] = \alpha \mathcal{O}[x_1(t)] + \beta \mathcal{O}[x_2(t)] = \alpha y_1(t) + \beta y_2(t). \tag{3.71}$$

- *Time invariance.* Delaying an incoming signal by some time $a > 0$ delays the outgoing signal by a with no other changes. Symbolically, that is

$$\mathcal{O}[x(t-a)] = y(t-a). \tag{3.72}$$

LTI systems are heavily constrained. So much so, we can actually derive a formula for $y(t)$ without specifying precisely what $\mathcal{O}$ is. To do this, we represent our incoming signal $x(t)$ as

$$x(t) = (x * \delta)(t) = \int_{0^-}^{\infty} x(\tau)\delta(t-\tau)\,\mathrm{d}\tau, \tag{3.73}$$

using the fact that the delta function is a unit for convolution. Now we apply $\mathcal{O}$ to $x(t)$, giving

$$\mathcal{O}[x(t)] = \mathcal{O}\left[\int_{0^-}^{\infty} x(\tau)\delta(t-\tau)\,\mathrm{d}\tau\right] = \int_{0^-}^{\infty} x(\tau)\mathcal{O}[\delta(t-\tau)]\,\mathrm{d}\tau, \tag{3.74}$$

where we've moved $\mathcal{O}$ into the integral because it is linear. Note that it does not act on $x(\tau)$ because it only acts on functions of t. In order to justify this step, we think of the integral as the limit of a sum and then assume $\mathcal{O}$ is suitably well-behaved so we can take this limit after introducing it. Writing $g(t) = \mathcal{O}[\delta(t)]$, we have that

$$y(t) = \int_{0^-}^{\infty} x(\tau)g(t-\tau)\,\mathrm{d}\tau, \tag{3.75}$$

where we've used the fact that the system is time invariant to write $\mathcal{O}[\delta(t-\tau)] = g(t-\tau)$. Therefore, the outgoing signal of an LTI system is given by the convolution of the incoming signal $x(t)$ with the function $g(t)$. We call $g(t)$ the *impulse response* since it measures how a system reacts to an incoming unit impulse.

When dealing with a system, it is natural to demand that the solution only depends on the present and past values of the incoming signal $x(t)$ since it wouldn't make much sense for $y(t)$ to know about future values of $x(t)$. This makes the system *causal*. However, since $y(t)$ is given as a convolution of $x(t)$ and $g(t)$, values of $x(\tau)$ for $\tau > t$ contribute to the integral, potentially violating causality. In order to avoid this, we need $g(t) = 0$ for $t < 0$, then

$$y(t) = \int_{0^-}^{t} x(\tau)g(t-\tau)\,\mathrm{d}\tau, \tag{3.76}$$

eliminating any contributions for $\tau > t$. The convolution now looks like the one we use when studying Laplace transforms, suggesting that we can use this to study LTI systems too.

We now restrict to the class of LTI systems whose outgoing signals are the solutions to a differential equation

$$a_n y^{(n)}(t) + a_{n-1} y^{(n-1)}(t) + \ldots + a_1 y'(t) + a_0 y(t) = x(t), \tag{3.77}$$

subject to the initial conditions

$$y^{(n-1)}(0) = y^{(n-2)}(0) = \ldots = y'(0) = y(0) = 0.$$

These particular initial conditions are required for the linearity of the system and indicate that the outgoing signal is totally at rest before we feed in an incoming signal. Due to these initial conditions, we have $\mathcal{L}[y^{(n)}(t)] = s^n \mathcal{L}[y(t)]$, so the Laplace transform of eq. (3.77) is

$$Y(s)(a_n s^n + a_{n-1} s^{n-1} + \ldots + a_1 s + a_0) = X(s),$$

which gives the solution

$$Y(s) = \frac{X(s)}{a_n s^n + a_{n-1} s^{n-1} + \ldots + a_1 s + a_0}. \tag{3.78}$$

By taking $x(t) = \delta(t)$, so $X(s) = 1$, we find that the Laplace transform of the impulse response is given by the rational function

$$G(s) = \frac{1}{a_n s^n + a_{n-1} s^{n-1} + \ldots + a_1 s + a_0}, \tag{3.79}$$

giving $Y(s) = G(s)X(s)$, as one would expect from the convolution theorem. The function $G(s)$ is called the *transfer function* of the system and we will see that understanding $G(s)$ alone is enough to qualitatively understand an LTI system.

Example 3.7. Consider the LTI system whose outgoing signal is the solution to

$$y''(t) + y(t) = x(t), \tag{3.80}$$

subject to the initial conditions $y'(0) = y(0) = 0$. Laplace transforming this equation, we find the solution for $Y(s)$

$$Y(s) = \frac{X(s)}{s^2 + 1}. \tag{3.81}$$

By taking $X(s) = 1$, we see that the transfer function is $1/(s^2+1)$, from which we obtain the impulse response $g(t) = \mathcal{L}^{-1}[1/(s^2+1)] = \sin t$. Now suppose the incoming signal is given by $x(t) = t$, then the outgoing signal is given by

$$\begin{aligned} y(t) = (x * g)(t) &= \int_0^t u \sin(t-u)\,\mathrm{d}u = [u\cos(t-u)]_0^t - \int_0^t \cos(t-u)\,\mathrm{d}u, \\ &= t + [\sin(t-u)]_0^t, \\ &= t - \sin t. \end{aligned} \tag{3.82}$$

We can easily check that $y(0) = y'(0) = 0$, as expected.

3.7 Stability of LTI systems

Given an LTI system, it is often instructive to figure out the long term behaviour of the system rather than explicitly solve for various incoming signals. In particular, we want to know whether an outgoing signal grows exponentially as $t \to \infty$ or if it remains finite and is *stable*. There are various ways of phrasing stability, so we give two different definitions for LTI systems.

Definition 3.4. (i) An LTI system is *stable* if

$$\lim_{t\to\infty} g(t) = 0, \tag{3.83}$$

where $g(t)$ is the impulse response.

(ii) The system is ***bounded-input, bounded-output (BIBO)*** stable if any bounded incoming signal results in a bounded outgoing signal. That is, for any incoming $x(t)$ satisfying $|x(t)| \leq A$ for all t, there exists $B \geq 0$ such that $|y(t)| \leq B$ for all t.

In many texts, BIBO stability is reformulated as requiring

$$\|g\|_1 = \int_0^\infty |g(t)|\,\mathrm{d}t < \infty, \tag{3.84}$$

meaning $g \in L^1(\mathbb{R}_{\geq 0})$. This is equivalent to our definition, since assuming $\|g\|_1 < \infty$ gives

$$|y(t)| = |(g * x)(t)| \leq \int_0^t |g(\tau)||x(t-\tau)|\,\mathrm{d}\tau \leq A\int_0^t |g(\tau)|\,\mathrm{d}\tau, \tag{3.85}$$

where we're using $|x(t)| \leq A$ since we assume bounded input. Taking $g \in L^1(\mathbb{R}_{\geq 0})$ ensures that the right-hand side is less than or equal to $A\|g\|_1$, thus bounding the output.

For general LTI systems, BIBO stability and regular stability are different notions. The first notion of stability only requires that the system's response to a unit impulse eventually dies out, but it doesn't say anything for more general incoming signals. To see this, consider the LTI system with impulse response $1/(t+1)$, then $g(t) \to 0$ as $t \to \infty$ so the system is stable in the first sense. However, if we supply the bounded signal $x(t) = 1$ the solution is

$$y(t) = (g * x)(t) = \int_0^t \frac{1}{u+1}\,\mathrm{d}u = \log(1+t), \tag{3.86}$$

which is unbounded and shows that the system is not BIBO stable. Thankfully, for causal LTI systems with rational transfer functions the two notions are equivalent and we have the following result.

Theorem 3.4. A causal LTI system with rational transfer function $G(s) = P(s)/Q(s)$ is stable if and only if the poles of $G(s)$ belong to the half-plane $\mathbb{H} = \{s : \mathrm{Re}(s) < 0\}$.

Recall that the poles of $G(s)$ are the values of s for which $G(s) = \infty$. If $P(s)$ and $Q(s)$ do not share any roots, then these are just the values of s for which $Q(s) = 0$. The proof of this theorem is non-examinable, but the general idea is to use partial fractions to write $G(s)$ as Laplace transforms of functions $t^n \mathrm{e}^{\alpha t}$ and then justify when $g(t)$ tends to zero.

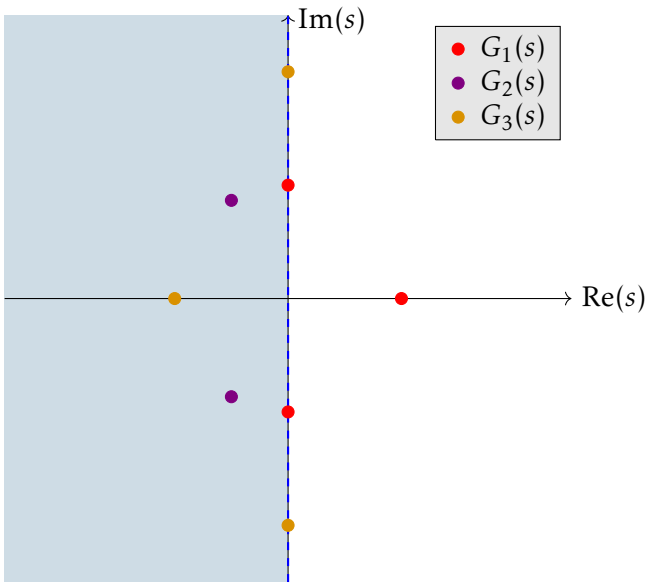


Figure 7: The poles of the transfer functions in example 3.8. The stable region $\mathrm{Re}(s) < 0$ is shaded and the boundary highlighted with a dashed blue line.

Example 3.8. Consider the stability of the systems with the following transfer functions

$$\text{(i)}\ \ G_1(s) = \frac{1}{(s-1)(s^2+1)}, \quad \text{(ii)}\ \ G_2(s) = \frac{s}{s^2+s+1}, \quad \text{(iii)}\ \ G_3(s) = \frac{s^2-s}{s^3+s^2+4s+4}. \tag{3.87}$$

(i) The poles of $G_1(s)$ occur when $s^2+1=0$ or $s-1=0$, so the roots are $s_\pm = \pm\mathrm{i}$ and $s=1$. These have real part 0 and 1 respectively, meaning the system is not stable.

(ii) The fact that the numerator isn't constant means that we should check if $P(s)$ and $Q(s)$ share any roots and reduce the fraction. This isn't the case here, as $Q(0)=1$, so the poles occur when

$$s^2+s+1=0 \Longrightarrow \left(s+\frac{1}{2}\right)^2+\frac{3}{4}=0, \tag{3.88}$$

giving poles $s_\pm = -\frac{1}{2} \pm \mathrm{i}\frac{\sqrt{3}}{2}$. Since $\mathrm{Re}(s_\pm) = -\frac{1}{2}$, this system is stable.

(iii) The zeroes of $P(s)$ are 0 and 1, but none of these have $Q(s)=0$ so we don't need to reduce the fraction. By inspection we see that $Q(-1)=0$, so $s_1=-1$ is a pole. This leads to the factorisation

$$s^3+s^2+4s+4=(s+1)(s^2+4), \tag{3.89}$$

from which we see that the other two poles are $s_\pm = \pm 2\mathrm{i}$. These have zero real part, so this system isn't stable either.

4 Partial differential equations

4.1 The heat equation

Thus far we have considered differential equations involving functions of one variable, but in reality we often have functions that depend on many variables. In this chapter we will study differential equations for functions of two variables $(x,t) \in D \times \mathbb{R}_{\geq 0}$ where $D \subseteq \mathbb{R}$. We will often use the notation

$$u_{xx}(x,t) = \frac{\partial^2 u}{\partial x^2}, \quad u_{xt}(x,t) = \frac{\partial^2 u}{\partial x \partial t}, \quad u_{tt}(x,t) = \frac{\partial^2 u}{\partial t^2}, \tag{4.1}$$

and so on to denote the partial derivatives of a function $u : D \times \mathbb{R}_{\geq 0} \to \mathbb{R}$.

The first partial differential equation (PDE) we will study is the *heat equation*. A function $u : [0,L] \times \mathbb{R}_{\geq 0} \to \mathbb{R}$ satisfies the heat equation if

$$u_t(x,t) = \kappa u_{xx}(x,t), \tag{4.2}$$

where κ is the diffusivity constant. Interpreting $u(x,t)$ as a temperature profile, κ measures how easily heat spreads throughout the domain, with large values of κ meaning that heat conducts more easily. When specifying any PDE, we must also give the boundary conditions and initial conditions in order to fully solve them. The boundary conditions of a PDE tell us about the behaviour of $u(x,t)$ at the boundaries and we will focus on two important classes of boundary conditions: *Dirichlet* and *Neumann* conditions.

Dirichlet	Neumann
$u(0,t) = \alpha,\ u(L,t) = \beta$	$u_x(0,t) = \alpha,\ u_x(L,t) = \beta$

Dirichlet conditions fix the value of $u(x,t)$ at the boundary points, while Neumann conditions allow the temperature profile to change at the boundary as long as the x-partial derivatives are maintained there. If $\alpha = \beta = 0$ in the above conditions, we say that these are *homogeneous conditions*. Homogeneous boundary conditions are nice, since if $u(x,t)$ satisfies the boundary conditions, then so does $\lambda u(x,t)$ for any $\lambda \in \mathbb{R}$. In addition to the boundary conditions, we will specify the initial conditions by fixing $u(x,0)$ as

$$u(x,0) = f(x), \quad 0 \leq x \leq L, \tag{4.3}$$

for a given function $f(x)$.

Solution with Dirichlet boundary conditions. Let us solve the heat equation with homogeneous Dirichlet boundary conditions and initial condition specified as

$$u(0,t) = u(L,t) = 0, \quad u(x,0) = f(x). \tag{4.4}$$

Generally, solutions to PDEs can display intricate dependency on both x and t, but for particularly nice PDEs we can decouple the x and t dependence and use the ansatz

$$u(x,t) = X(x)T(t). \tag{4.5}$$

This is called the *separation of variables* ansatz. Substituting this into the heat equation gives

$$X(x)T'(t) = \kappa X''(x)T(t), \tag{4.6}$$

which upon dividing by $X(x)T(t)$ gives

$$\frac{T'(t)}{T(t)} = \kappa \frac{X''(x)}{X(x)}. \tag{4.7}$$

Notice that the left-hand side of this equation only depends on t, while the right-hand side only depends on x. In order for this to be the case, we must have that both sides are actually constant! If we call this constant λ, we get two ordinary differential equations

$$T'(t) = \lambda T(t), \quad X''(x) = \frac{\lambda}{\kappa} X(x), \tag{4.8}$$

which we can easily solve. Focussing first on the equation for $X(x)$, we need to be careful about the sign of λ/κ as it will change whether exponentials or trigonometric functions will appear in the solution.

- Taking $\lambda/\kappa = \alpha^2 > 0$, we can easily solve for $X(x)$ as

  $$X(x) = A\mathrm{e}^{\alpha x} + B\mathrm{e}^{-\alpha x}. \tag{4.9}$$

  Imposing $u(0,t) = 0$ gives $A = -B$, while imposing $u(L,t) = 0$ gives

  $$0 = A(\mathrm{e}^{\alpha L} - \mathrm{e}^{-\alpha L}). \tag{4.10}$$

  Since $\mathrm{e}^{\alpha L} - \mathrm{e}^{-\alpha L} = 0$ implies that $L = 0$, we disregard this possibility and take $A = 0$. In other words, if $\lambda/\kappa > 0$ then $u(x,t) = 0$ for all values of x and t.

- If $\lambda/\kappa = 0$, then integrating $X''(x) = 0$ twice gives the solution $X(x) = Ax + B$. However, imposing the boundary conditions gives $A = B = 0$ again, so $u(x,t) = 0$ for all x and t.

- Now writing $\lambda/\kappa = -\alpha^2 < 0$, we have

  $$X(x) = A\cos\alpha x + B\sin\alpha x. \tag{4.11}$$

  The condition $u(0,t) = 0$ tells us that $A = 0$, while $u(L,t) = 0$ gives

  $$B\sin\alpha L = 0. \tag{4.12}$$

  As we're looking for non-trivial solutions, we disregard the possibility that $B = 0$, so we must impose $\sin\alpha L = 0$. This means that $\alpha L = \pi n$ for some integer n, so

  $$\alpha = \frac{n\pi}{L}, \quad n \in \mathbb{N}. \tag{4.13}$$

  Therefore, we have a solution $X(x)$ for each value of n given by

  $$X_n(x) = b_n \sin\left(\frac{n\pi x}{L}\right). \tag{4.14}$$

Having found the solutions $X_n(x)$ for $\lambda/\kappa = -\alpha^2 < 0$, we now need to find the corresponding solutions for $T(t)$. As the equation for $T(t)$ is a first order differential equation, the solution is given by

$$T_n'(t) = -\alpha^2 \kappa T_n(t) \Longrightarrow T_n(t) = C\mathrm{e}^{-\alpha^2 \kappa t}, \tag{4.15}$$

where $\alpha = n\pi/L$. Therefore, the function

$$u_n(x,t) = b_n \mathrm{e}^{-\kappa n^2 \pi^2 t/L^2} \sin\left(\frac{n\pi x}{L}\right), \tag{4.16}$$

solves the heat equation and satisfies the boundary conditions for any $n \in \mathbb{N}$. However, we've neglected the initial condition $u(x,0) = f(x)$, so this ***isn't*** a solution to the full problem.

In order to implement the initial condition, we note that the heat equation with homogeneous boundary conditions is ***linear***, meaning that the sum of two solutions is also a solution. As such, a general solution can be written as

$$u(x,t) = \sum_{n=1}^{\infty} b_n \mathrm{e}^{-\kappa n^2 \pi^2 t/L^2} \sin\left(\frac{n\pi x}{L}\right). \tag{4.17}$$

Setting $t = 0$ and equating this to the initial configuration $f(x)$ imposes

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right). \tag{4.18}$$

This is familiar to us— it is the sine half-range series for $f(x)$! We can therefore extend $f(x)$ to an odd function on $[-L, L]$ and use the formulae from section 1.4 to find b_n as

$$b_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) \mathrm{d}x. \tag{4.19}$$

In other words, solving the heat equation with homogeneous Dirichlet boundary conditions boils down to finding a Fourier sine series for the initial configuration.

Example 4.1. Suppose $u : [0,\pi] \times \mathbb{R}_{\geq 0} \to \mathbb{R}$ satisfies the heat equation with Dirichlet boundary conditions

$$u(0,t) = u(\pi,t) = 0, \tag{4.20}$$

and initial condition

$$u(x,0) = x(\pi - x). \tag{4.21}$$

As shown above, the general solution is of the form

$$u(x,t) = \sum_{n=1}^{\infty} b_n \mathrm{e}^{-\kappa n^2 t} \sin nx, \tag{4.22}$$

with coefficients given by the Fourier half-range series formula

$$\begin{aligned} b_n &= \frac{2}{\pi}\int_0^\pi x(\pi - x)\sin nx\,\mathrm{d}x, \\ &= \frac{2}{n\pi}[-x(\pi-x)\cos nx]_0^\pi + \frac{2}{n\pi}\int_0^\pi (\pi - 2x)\cos nx\,\mathrm{d}x, \\ &= \frac{2}{n^2\pi}[(\pi-2x)\sin nx]_0^\pi + \frac{4}{n^2\pi}\int_0^\pi \sin nx\,\mathrm{d}x, \\ &= \frac{4}{n^3\pi}[-\cos nx]_0^\pi = \frac{4(1-(-1)^n)}{\pi n^3}. \end{aligned} \tag{4.23}$$

When n is even, $b_n = 0$, so letting $n = 2j+1$ lets us write the solution as

$$u(x,t) = \frac{8}{\pi}\sum_{j=0}^{\infty} \frac{1}{(2j+1)^3} \mathrm{e}^{-\kappa(2j+1)^2 t}\sin(2j+1)x. \tag{4.24}$$

Solution with Neumann boundary conditions. Suppose we now impose the homogeneous Neumann boundary conditions

$$u_x(0,t) = u_x(L,t) = 0, \quad u(x,0) = f(x). \tag{4.25}$$

Of course, the heat equation itself hasn't changed, so using separation of variables $u(x,t) = X(x)T(t)$ we get the same equations

$$X''(x) = \frac{\lambda}{\kappa}X(x), \quad T'(t) = \lambda T(t). \tag{4.26}$$

Again, the behaviour of the solution depends on the value of λ/κ and we need to treat these cases separately when imposing the boundary conditions.

- For $\lambda/\kappa = \alpha^2 > 0$, the solution is given by a sum of exponentials

  $$X(x) = A\mathrm{e}^{\alpha x} + B\mathrm{e}^{-\alpha x}. \tag{4.27}$$

  As the Neumann conditions involve the derivatives $u_x(x,t) = X'(x)T(t)$, we impose $X'(0) = X'(L) = 0$ where

  $$X'(x) = A\alpha\mathrm{e}^{\alpha x} - B\alpha\mathrm{e}^{-\alpha x}. \tag{4.28}$$

  Substituting in $x = 0$ and $x = L$ readily shows that $A = B = 0$, giving $u(x,t) = 0$ for all x and t.

- When $\lambda/\kappa = 0$ we get $X(x) = Ax + B$ and $X'(x) = A$. Imposing the boundary conditions gives $A = 0$, but doesn't fix B. Therefore, we get a possible solution with $X(x)$ constant. However, since the general solution will not depend on x at all, we can only have this solution if $u(x,0)$ is also constant.

- Finally, when $\lambda/\kappa = -\alpha^2 < 0$ we obtain the trigonometric solution

$$X(x) = A\cos\alpha x + B\sin\alpha x \Longrightarrow X'(x) = -A\alpha\sin\alpha x + B\alpha\cos\alpha x. \tag{4.29}$$

The condition $u_x(0,t) = 0$ tells us that $B = 0$, while $u_x(L,t) = 0$ gives

$$A\alpha\sin\alpha L = 0 \Longrightarrow \alpha = \frac{n\pi}{L}, \tag{4.30}$$

for $n \in \mathbb{Z}_{\geq 0}$. Therefore, we get the solutions

$$X_n(x) = a_n\cos\left(\frac{n\pi x}{L}\right). \tag{4.31}$$

Thus far, the only difference to the Dirichlet case is that we obtain solutions with cosines instead of sines.

Solving for $T(t)$ is exactly the same as before, therefore the general solution takes the form

$$u(x,t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n e^{-\kappa n^2\pi^2 t/L^2}\cos\left(\frac{n\pi x}{L}\right) \tag{4.32}$$

Imposing the initial condition $u(x,0) = f(x)$ now gives

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n\cos\left(\frac{n\pi x}{L}\right), \tag{4.33}$$

and we recognise this as the cosine half-range series for $f(x)$. As in the Dirichlet case, we obtain the expressions for a_n by using the half-range formulae

$$a_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)\mathrm{d}x. \tag{4.34}$$

Overall, in both the Dirichlet and Neumann case we obtain a series expansion that depends exponentially on t but trigonometrically on x with the only difference being the type of Fourier half-range series that appears for each type of boundary condition. Therefore, for the heat equation we have the motto that 'Dirichlet = sine series' and 'Neumann = cosine series'.

Example 4.2. Suppose $u : [0,1]\times\mathbb{R}_{\geq 0} \to \mathbb{R}$ satisfies the heat equation with Neumann boundary conditions

$$u_x(0,t) = u_x(1,t) = 0, \tag{4.35}$$

and initial condition

$$u(x,0) = x. \tag{4.36}$$

The general solution takes the form

$$u(x,t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n e^{-\kappa n^2\pi^2 t}\cos(n\pi x), \tag{4.37}$$

where the coefficients for $n > 0$ are given by

$$\begin{aligned} a_n &= 2\int_0^1 x\cos(n\pi x)\,\mathrm{d}x, \\ &= \frac{2}{n\pi}[x\sin(n\pi x)]_0^1 - \frac{2}{n\pi}\int_0^1 \sin(n\pi x)\,\mathrm{d}x, \\ &= \frac{2}{(n\pi)^2}[\cos(n\pi x)]_0^1 = \frac{2((-1)^n-1)}{(n\pi)^2}, \end{aligned} \tag{4.38}$$

which is zero when n is even. Since this is singular when $n = 0$, we need to find a_0 separately. Explicitly, we have

$$a_0 = 2\int_0^1 x\,\mathrm{d}x = 1. \tag{4.39}$$

Writing $n = 2j + 1$, the solution is given by

$$u(x,t) = \frac{1}{2} - \frac{4}{\pi^2}\sum_{j=0}^{\infty}\frac{1}{(2j+1)^2}\mathrm{e}^{-\kappa(2j+1)^2\pi^2 t}\cos(2j+1)\pi x. \tag{4.40}$$

4.2 The wave equation

The second PDE we will deal with is the *wave equation* that describes the propagation of waves through a medium. We'll focus on the 1-dimensional case where the function $u : [0, L]\times\mathbb{R}_{\geq 0} \to \mathbb{R}$ satisfies

$$u_{tt}(x,t) = c^2 u_{xx}(x,t). \tag{4.41}$$

Here we can think of $u(x,t)$ as the height of a wave over a point x at a time t and the constant c describes the propagation speed of the wave. We will also imagine that the wave is fixed at the boundary of the interval $[0, L]$ at the same level, therefore restricting us to homogeneous Dirichlet boundary conditions

$$u(0,t) = u(L,t) = 0. \tag{4.42}$$

Additionally, as there is an additional t-derivative when compared with the heat equation we need more initial conditions. Therefore, we must specify

$$u(x,0) = f(x), \quad u_t(x,0) = g(x). \tag{4.43}$$

In other words, we need to specify both the initial profile $f(x)$ of the wave along with its initial velocity profile $g(x)$.

We proceed as before by using separation of variables to write $u(x,t) = X(x)T(t)$, giving

$$X(x)T''(t) = c^2 X''(x)T(t), \tag{4.44}$$

which, upon division by $X(x)T(t)$, we get

$$\frac{T''(t)}{T(t)} = c^2 \frac{X''(x)}{X(x)}. \tag{4.45}$$

Since the left-hand side depends only on t, while the right-hand side depends only on x, these must be equal to a constant λ. Hence

$$T''(t) = \lambda T(t), \quad X''(x) = \frac{\lambda}{c^2} X(x). \tag{4.46}$$

The only difference when compared to the heat equation is that the equation for $T(t)$ is a second order differential equation, rather than first order. However, solving for $X(x)$ is completely analogous to the heat equation: we consider the cases $\lambda/c^2 < 0$, $\lambda = 0$ and $\lambda/c^2 > 0$ separately and come to the conclusion that only the $\lambda/c^2 < 0$ case is important. Restricting to this case, we get

$$X(x) = A\cos\alpha x + B\sin\alpha x, \tag{4.47}$$

where $\lambda/c^2 = -\alpha^2$. The boundary condition $u(0,t) = 0$ gives us $A = 0$, while $u(L,t) = 0$ constrains α to

$$\alpha = \frac{n\pi}{L}, \quad n \in \mathbb{N}. \tag{4.48}$$

Fixing the value of α, the corresponding solution for $T(t)$ is then

$$T_n(t) = C_n \cos\left(\frac{n\pi ct}{L}\right) + D_n \sin\left(\frac{n\pi ct}{L}\right). \tag{4.49}$$

Unlike the heat equation, we now get trigonometric dependence on both t and x. The wave equation with homogeneous boundary conditions is linear, so we can sum our solutions together as before. This gives the general solution

$$u(x,t) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{L}\right)\left(C_n \cos\left(\frac{n\pi ct}{L}\right) + D_n \sin\left(\frac{n\pi ct}{L}\right)\right). \tag{4.50}$$

Finally, we need to apply the initial conditions, both to $u(x,0)$ and the velocity profile $u_t(x,0)$. Using the initial condition $u(x,0) = f(x)$, we have

$$f(x) = \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi x}{L}\right), \tag{4.51}$$

realising $f(x)$ as a Fourier sine series. Therefore, the coefficients C_n are

$$C_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)\mathrm{d}x. \tag{4.52}$$

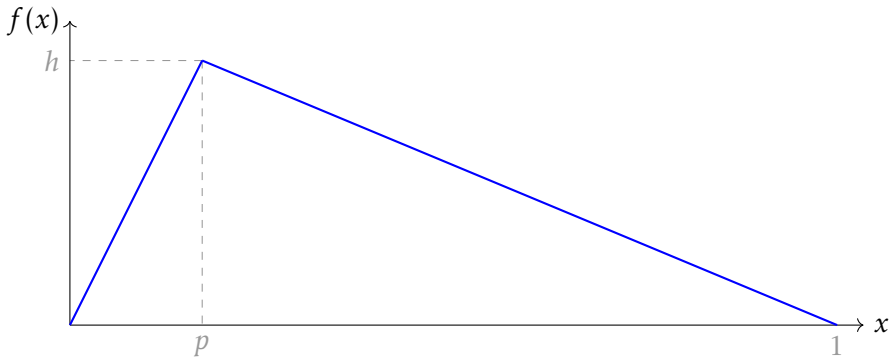


Figure 8: The initial configuration of the plucked string.

Applying the initial condition $u_t(x,0)$ gives

$$g(x) = \sum_{n=1}^{\infty} D_n \left(\frac{n\pi c}{L}\right) \sin\left(\frac{n\pi x}{L}\right). \tag{4.53}$$

Again, this is a Fourier sine series, but we should be careful when using the coefficient formula to include the $n\pi c/L$ factor. Doing so gives

$$D_n\left(\frac{n\pi c}{L}\right) = \frac{2}{L}\int_0^L g(x)\sin\left(\frac{n\pi x}{L}\right)\mathrm{d}x, \tag{4.54}$$

and dividing by $n\pi c/L$ gives the coefficients

$$D_n = \frac{2}{n\pi c}\int_0^L g(x)\sin\left(\frac{n\pi x}{L}\right)\mathrm{d}x. \tag{4.55}$$

All in all, solving the wave equation with homogeneous Dirichlet boundary conditions reduces to finding the Fourier sine series for the initial configuration $f(x)$ and the initial velocity profile $g(x)$, with an additional factor $L/n\pi c$ included in the coefficients for the latter.

Example 4.3. We want to model a string that is plucked to a height h at some point p along its length. By taking the string to lie on the interval $[0,1]$, the initial configuration is given by (drawn in fig. 8)

$$f(x) = \begin{cases} hx/p, & 0 \le x \le p, \\ h(1-x)/(1-p), & p < x \le 1. \end{cases} \tag{4.56}$$

We will assume that the string is at rest as it is plucked, so $u_t(x,0) = 0$. Using this in the equation for the D_n coefficients, we see that $D_n = 0$ for every n. The solution is therefore of the form

$$u(x,t) = \sum_{n=1}^{\infty} C_n \sin(n\pi x)\cos(n\pi c t). \tag{4.57}$$

Now we use the Fourier sine series formula to get

$$C_n = 2\int_0^1 f(x)\sin(n\pi x)\,\mathrm{d}x = \frac{2h}{p}\int_0^p x\sin(n\pi x)\,\mathrm{d}x + \frac{2h}{1-p}\int_p^1 (1-x)\sin(n\pi x)\,\mathrm{d}x. \tag{4.58}$$

Call the first integral on the right-hand side I_1 and the second I_2. Using integration by parts, we calculate I_1 as

$$\begin{aligned} I_1 &= \frac{2h}{p}\int_0^p x\sin(n\pi x)\,\mathrm{d}x, \\ &= \frac{2h}{n\pi p}\left[-x\cos(n\pi x)\right]_0^p + \frac{2h}{n\pi p}\int_0^p \cos(n\pi x)\,\mathrm{d}x, \\ &= -\frac{2h}{n\pi}\cos(n\pi p) + \frac{2h}{(n\pi)^2 p}\left[\sin(n\pi x)\right]_0^p = \frac{2h}{(n\pi)^2 p}\sin(n\pi p) - \frac{2h}{n\pi}\cos(n\pi p). \end{aligned} \tag{4.59}$$

Similarly, integration by parts on I_2 gives

$$\begin{aligned} I_2 &= \frac{2h}{1-p}\int_p^1 (1-x)\sin(n\pi x)\,\mathrm{d}x, \\ &= \frac{2h}{n\pi(1-p)}\left[-(1-x)\cos(n\pi x)\right]_p^1 - \frac{2h}{n\pi(1-p)}\int_p^1 \cos(n\pi x)\,\mathrm{d}x, \\ &= \frac{2h}{n\pi}\cos(n\pi p) - \frac{2h}{(n\pi)^2(1-p)}\left[\sin(n\pi x)\right]_p^1, \\ &= \frac{2h}{(n\pi)^2(1-p)}\sin(n\pi p) + \frac{2h}{n\pi}\cos(n\pi p). \end{aligned} \tag{4.60}$$

When we add I_1 and I_2 the cosine term cancels, so we arrive at

$$\begin{aligned} C_n &= \frac{2h}{(n\pi)^2 p}\sin(n\pi p) + \frac{2h}{(n\pi)^2(1-p)}\sin(n\pi p), \\ &= \frac{2h\sin(n\pi p)}{p(1-p)(n\pi)^2}. \end{aligned} \tag{4.61}$$

Now that we've done the painful work, we have that the solution to the wave equation with the plucked string initial conditions is given by

$$u(x,t) = \frac{2h}{p(1-p)\pi^2}\sum_{n=1}^{\infty}\left(\frac{\sin(n\pi p)}{n^2}\right)\sin(n\pi x)\cos(n\pi c t). \tag{4.62}$$

Choosing some sensible values for h, c and p and plotting this for various values of t shows that the wave propagates away from the plucked point until it hits a boundary, thereafter it is reflected back towards the other direction. Compare this with this slow motion video of a plucked string.

4.3 Solving inhomogeneous PDEs

Suppose we are given a partial differential equation together with some boundary conditions. We say that it is *inhomogeneous* if there exist solutions $u(x,t)$ and $v(x,t)$ satisfying the boundary conditions such that a linear combination $\alpha u(x,t) + \beta v(x,t)$ does not satisfy the PDE with specified boundary conditions. For example, consider the *forced heat equation*

$$u_t(x,t) - \kappa u_{xx}(x,t) = f(x), \tag{4.63}$$

satisfying $u(0,t) = u(L,t) = 0$. If $u(x,t)$ and $v(x,t)$ are solutions, then $u(x,t) + v(x,t)$ still satisfies the boundary conditions, but we have that

$$\frac{\partial}{\partial t}(u+v) - \kappa \frac{\partial^2}{\partial x^2}(u+v) = (u_t - \kappa u_{xx}) + (v_t - \kappa v_{xx}). \tag{4.64}$$

Each of the bracketed terms is just the left-hand side of eq. (4.63), so we have

$$(u+v)_t - \kappa(u+v)_{xx} = 2f(x). \tag{4.65}$$

Therefore, $u(x,t) + v(x,t)$ is ***not*** a solution to the PDE. Generally, given a linear PDE, adding a term independent of the solution $u(x,t)$ breaks the homogeneity of the system. Here the forcing term $f(x)$ is responsible for this.

A second way homogeneity can fail is through its boundary conditions. If we now consider the heat equation with boundary conditions $u(0,t) = u(L,t) = \pi$, then the sum of two solutions $u(x,t)$ and $v(x,t)$ satisfies the boundary conditions

$$(u+v)(0,t) = (u+v)(L,t) = 2\pi. \tag{4.66}$$

These are the wrong boundary conditions! For Dirichlet and Neumann boundary conditions, the system can be homogeneous only if $u(0,t) = u(L,t) = 0$ or $u_x(0,t) = u_x(L,t) = 0$.

So how can we solve eq. (4.63) with general inhomogeneous boundary conditions $u(0,t) = \alpha$ and $u(L,t) = \beta$? We can do so by writing $u(x,t)$ as

$$u(x,t) = v(x,t) + w(x), \tag{4.67}$$

where $v(x,t)$ satisfies the homogeneous heat equation

$$v_t(x,t) - \kappa v_{xx}(x,t) = 0, \tag{4.68}$$

with boundary conditions $v(0,t) = v(L,t) = 0$. The function $w(x)$ is called a ***stationary solution***, as it does not depend on t. Its purpose is to adjust the homogeneous solution so we can satisfy both the forcing and the boundary conditions. Substituting $u(x,t)$ into the left-hand side of the forced heat equation then gives

$$(v(x,t) + w(x))_t - \kappa(v(x,t) + w(x))_{xx} = (v_t(x,t) - \kappa v_{xx}(x,t)) - \kappa w''(x). \tag{4.69}$$

By hypothesis, the term in the bracket vanishes, so we get the condition

$$w''(x) = -\frac{f(x)}{\kappa}. \tag{4.70}$$

Imposing the inhomogeneous boundary conditions also implies that

$$w(0) = \alpha, \quad w(L) = \beta. \tag{4.71}$$

Therefore, in order to solve the forced heat equation, we need to solve the second order ODE eq. (4.70) subject to the boundary conditions above. The only caveat occurs when considering the initial condition

$$u(x,0) = g(x) \Longrightarrow v(x,0) = g(x) - w(x). \tag{4.72}$$

Therefore, the full solution is given by

$$u(x,t) = w(x) + \sum_{n=1}^{\infty} b_n \mathrm{e}^{-\kappa n^2 \pi^2 t/L^2} \sin\left(\frac{\pi n x}{L}\right), \tag{4.73}$$

where the coefficients are given by

$$b_n = \frac{2}{L}\int_0^L (g(x) - w(x)) \sin\left(\frac{\pi n x}{L}\right) \mathrm{d}x. \tag{4.74}$$

This idea generalises to other inhomogeneous PDEs: split up $u(x,t)$ as $v(x,t)+w(x)$ with $v(x,t)$ satisfying the homogeneous version of the problem, substitute this into the equation to find an ODE for $w(x)$ and solve it. Then rewrite the initial condition for $v(x,t)$ and calculate the appropriate Fourier coefficients.

Example 4.4. For the first example, consider the unforced heat equation

$$u_t(x,t) = \kappa u_{xx}(x,t), \tag{4.75}$$

with the boundary and initial conditions

$$u(0,t) = 0, \quad u(\pi,t) = -\pi^2, \quad u(x,0) = -x^2. \tag{4.76}$$

First, write $u(x,t) = v(x,t) + w(x)$ with $v(x,t)$ the solution to the homogeneous problem and $w(x)$ a stationary solution. Substituting this into the heat equation yields

$$w''(x) = 0 \Longrightarrow w(x) = Ax + B, \tag{4.77}$$

where A and B are constants of integration. The boundary condition $u(0,t) = 0$ implies $B = 0$, while the other boundary condition gives

$$A\pi = -\pi^2 \Longrightarrow w(x) = -\pi x. \tag{4.78}$$

The solution is therefore of the form

$$u(x,t) = -\pi x + \sum_{n=1}^{\infty} b_n \mathrm{e}^{-\kappa n^2 t} \sin nx. \tag{4.79}$$

The coefficients can be found by using the initial condition for $v(x,t)$ given by

$$v(x,0) = u(x,0) - w(x) = \pi x - x^2 = x(\pi - x), \tag{4.80}$$

hence

$$b_n = \frac{2}{\pi}\int_0^\pi x(\pi - x)\sin nx\,\mathrm{d}x. \tag{4.81}$$

Thankfully, we calculated these integrals in example 4.1, so we can conclude that the solution is

$$u(x,t) = -\pi x + \frac{8}{\pi}\sum_{j=0}^{\infty}\frac{1}{(2j+1)^3}\mathrm{e}^{-\kappa(2j+1)^2 t}\sin(2j+1)x. \tag{4.82}$$

Example 4.5. Consider the forced wave equation

$$u_{tt}(x,t) - u_{xx}(x,t) = \sin\pi x, \tag{4.83}$$

with boundary and initial conditions

$$u(0,t) = u(1,t) = 0, \quad u(x,0) = 0, \quad u_t(x,0) = \sin 3\pi x. \tag{4.84}$$

Decompose the solution as $u(x,t) = v(x,t) + w(x)$ with $v(x,t)$ the homogeneous solution. Substituting this into the forced wave equation gives

$$w''(x) = -\sin\pi x. \tag{4.85}$$

Integrating this twice gives the stationary solution

$$w(x) = \frac{1}{\pi^2}\sin\pi x + Ax + B \tag{4.86}$$

Since $\sin 0 = \sin\pi = 0$, the boundary conditions impose $A = B = 0$. Therefore, the solution takes the form

$$u(x,t) = \frac{1}{\pi^2}\sin\pi x + \sum_{n=1}^{\infty}\sin(n\pi x)\left(C_n\cos(n\pi t) + D_n\sin(n\pi t)\right). \tag{4.87}$$

The initial condition for $v(x,t)$ is $v(x,0) = -w(x)$, so the C_n coefficients are given by

$$C_n = -\frac{2}{\pi^2}\int_0^1 \sin\pi x\sin n\pi x\,\mathrm{d}x = \begin{cases} -1/\pi^2, & n = 1, \\ 0, & \text{otherwise.} \end{cases} \tag{4.88}$$

This can be justified by using the product formula for sines or by expanding each term into complex exponentials. As $w(x)$ does not depend on t, the initial condition for $v_t(x,t)$ is the

same as $u_t(x,0) = \sin 3\pi x$. As such, the coefficient D_n is

$$D_n = \frac{2}{n\pi}\int_0^1 \sin 3\pi x \sin n\pi x \,\mathrm{d}x = \begin{cases} 1/3\pi, & n = 3, \\ 0, & \text{otherwise.} \end{cases} \tag{4.89}$$

Therefore, only C_1 and D_3 are non-zero. As such, the full solution is given by

$$u(x,t) = \frac{1}{\pi^2}\sin \pi x - \frac{1}{\pi^2}\sin \pi x \cos \pi t + \frac{1}{3\pi}\sin 3\pi x \sin 3\pi t. \tag{4.90}$$

4.4 The heat equation on an infinite domain ★

Just as Fourier series can be used to solve the heat equation on an interval $[0, L]$, the Fourier transform can help us solve it over the whole real line $\mathbb{R}$. Concretely, suppose $u : \mathbb{R} \times \mathbb{R}_{\geq 0} \to \mathbb{R}$ satisfies the heat equation with initial condition

$$u(x,0) = f(x), \tag{4.91}$$

where $f \in L^1(\mathbb{R})$. We will assume that $u(x,t) \in L^1(\mathbb{R})$ for any fixed value of t, so $\lim_{x\to\pm\infty} u(x,t) = 0$. By assuming this, we can define the Fourier transform of $u(x,t)$ with respect to the x variable as usual

$$\hat{u}(\xi,t) = \int_{-\infty}^{\infty} u(x,t)\mathrm{e}^{-\mathrm{i}\xi x}\,\mathrm{d}x. \tag{4.92}$$

Taking the Fourier transform of the heat equation then gives

$$\hat{u}_t(\xi,t) = -\kappa\xi^2\hat{u}(\xi,t). \tag{4.93}$$

This is a first order differential equation for $\hat{u}(\xi,t)$ that we can solve easily by rearranging and integrating as

$$\int \frac{1}{\hat{u}}\frac{\partial \hat{u}}{\partial t}\,\mathrm{d}t = -\kappa\int \xi^2\,\mathrm{d}t \implies \log \hat{u}(\xi,t) = C(\xi) - \kappa\xi^2 t. \tag{4.94}$$

Here the constant of integration $C(\xi)$ can depend on ξ since we've integrated with respect to t. Defining $D(\xi) = \mathrm{e}^{C(\xi)}$, we can write the solution as

$$\hat{u}(\xi,t) = D(\xi)\mathrm{e}^{-\kappa\xi^2 t}. \tag{4.95}$$

The initial condition can also be Fourier transformed as $\hat{u}(\xi,0) = \hat{f}(\xi)$, hence

$$\hat{f}(\xi) = D(\xi). \tag{4.96}$$

Piecing this all together and using the Fourier inversion theorem brings us to the full solution

$$u(x,t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \hat{f}(\xi)\mathrm{e}^{\mathrm{i}\xi x - \kappa\xi^2 t}\,\mathrm{d}\xi. \tag{4.97}$$

Since this is an integral over ξ, one can check that this satisfies the heat equation by differentiating with respect to x and t under the integral sign.

As a simple example, consider the initial configuration $u(x,0) = \delta(x-y)$ for some point $y \in \mathbb{R}$. Then the Fourier transformed initial condition is

$$\hat{u}(\xi,0) = \int_{-\infty}^{\infty} \delta(x-y) \mathrm{e}^{-\mathrm{i}\xi x} \,\mathrm{d}x = \mathrm{e}^{-\mathrm{i}\xi y}, \tag{4.98}$$

and the full solution is

$$u(x,t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathrm{e}^{\mathrm{i}\xi(x-y)-\kappa\xi^2 t} \,\mathrm{d}\xi = \mathcal{F}^{-1}\left[\mathrm{e}^{-\mathrm{i}\xi y-\kappa\xi^2 t}\right]. \tag{4.99}$$

By using the transformation rules for Fourier transforms, we can deduce that

$$u(x,t) = \frac{1}{\sqrt{4\pi\kappa t}} \exp\left(-\frac{(x-y)^2}{4\kappa t}\right). \tag{4.100}$$

This is known as the *heat kernel*, a useful fundamental solution to the heat equation which has found usage in many fields of mathematics.

5 Discrete transforms

5.1 The $\mathcal{Z}$-transform

Integral transforms allow us to simplify problems involving functions of a continuous variable t, but in many practical situations we don't have access to a function defined for every value of t. Instead, we may only have knowledge of a function at certain time intervals and we have to adapt our techniques to deal with this discrete data.

The discrete analogue of functions are *sequences*, which we denote by $a = (a_n)_{n=0}^{\infty}$. If we wish to think of these as signals, we can think of the signal updating at even time intervals, with the value of the signal at a time $n\Delta t$ being a_n. Some simple examples are

$$(n)_{n=0}^{\infty} = (0,1,2,3,\ldots), \quad (\delta_n)_{n=0}^{\infty} = (1,0,0,0,\ldots), \quad (\lambda^n)_{n=0}^{\infty} = (1,\lambda,\lambda^2,\lambda^3,\ldots), \tag{5.1}$$

where $\lambda \in \mathbb{C}$ is some constant. Given two sequences $a = (a_n)_{n=0}^{\infty}$ and $b = (b_n)_{n=0}^{\infty}$, we can define linear combinations of them component-wise

$$\alpha a + \beta b = (\alpha a_n + \beta b_n)_{n=0}^{\infty} = (\alpha a_0 + \beta b_0, \alpha a_1 + \beta b_1, \ldots), \tag{5.2}$$

making the space of sequences indexed by $\mathbb{Z}_{\geq 0}$ a vector space.

Definition 5.1. Let $a = (a_n)_{n=0}^{\infty}$ be a sequence. The *$\mathcal{Z}$-transform* of a is given by

$$A(z) = \mathcal{Z}[a] = \sum_{n=0}^{\infty} a_n z^{-n}, \tag{5.3}$$

where z is a complex variable.

Much like the Laplace transform, the $\mathcal{Z}$-transform only really makes sense for values of z such that the summation converges. This defines the *region of convergence* of a given $\mathcal{Z}$-transform. In particular, if a sequence $(a_n)_{n=0}^{\infty}$ is bounded by

$$|a_n| \leq Ar^n, \quad A, r \geq 0, \tag{5.4}$$

making it an *exponential type* sequence, we have

$$|A(z)| \leq \sum_{n=0}^{\infty} |a_n||z^{-n}| \leq A \sum_{n=0}^{\infty} \left(\frac{r}{|z|}\right)^n. \tag{5.5}$$

If $r/|z| < 1$, the right-hand side converges and we can use the formula for geometric sums to get

$$|A(z)| \leq \frac{A}{1 - r/|z|} = \frac{A|z|}{|z| - r}. \tag{5.6}$$

Therefore, the region of convergence is given by $|z| > r$. In other examples, the region of convergence can often be inferred by using a known Maclaurin series or checking where the poles of the summed series lie.

Example 5.1. (i) For the sequence $(\lambda^n)_{n=0}^{\infty}$, the $\mathcal{Z}$-transform is given by

$$\mathcal{Z}[\lambda^n] = \sum_{n=0}^{\infty} \lambda^n z^{-n} = \sum_{n=0}^{\infty} \left(\frac{\lambda}{z}\right)^n. \tag{5.7}$$

Noting that the binomial expansion $1/(1-q) = 1+q+q^2+\ldots$ is valid for $|q|<1$, we can rewrite the above series as

$$\mathcal{Z}[\lambda^n] = \frac{1}{1-\lambda/z} = \frac{z}{z-\lambda}, \tag{5.8}$$

as long as $|z| > |\lambda|$.

(ii) Using the above, we can derive the $\mathcal{Z}$-transforms for $\cos \alpha n$ and $\sin \alpha n$. Concretely, we have

$$\mathcal{Z}[\cos \alpha n] = \frac{1}{2}(\mathcal{Z}[\mathrm{e}^{\mathrm{i}\alpha n}] + \mathcal{Z}[\mathrm{e}^{-\mathrm{i}\alpha n}]) = \frac{1}{2}\left(\frac{z}{z-\mathrm{e}^{\mathrm{i}\alpha}} + \frac{z}{z-\mathrm{e}^{-\mathrm{i}\alpha}}\right). \tag{5.9}$$

Combining the fractions, we can simplify this as

$$\mathcal{Z}[\cos \alpha n] = \frac{z^2 - \frac{1}{2}z(\mathrm{e}^{\mathrm{i}\alpha} + \mathrm{e}^{-\mathrm{i}\alpha})}{z^2 - z(\mathrm{e}^{\mathrm{i}\alpha} + \mathrm{e}^{-\mathrm{i}\alpha}) + 1} = \frac{z^2 - z\cos\alpha}{z^2 - 2z\cos\alpha + 1}. \tag{5.10}$$

Similarly, the $\mathcal{Z}$-transform of $\sin \alpha n$ is

$$\mathcal{Z}[\sin \alpha n] = \frac{z\sin\alpha}{z^2 - 2z\cos\alpha + 1}. \tag{5.11}$$

The poles of these functions occur at $z_{\pm} = \mathrm{e}^{\pm \mathrm{i}\alpha}$, so the region of convergence is given by $|z| > 1$.

5.2 Relation to Laplace transform ★

The $\mathcal{Z}$-transform can be thought of as a discrete version of the Laplace transform. To see this, we first define the *Dirac comb* $\text{Ш}_T(t)$[2] to be

$$\text{Ш}_T(t) = \sum_{n=0}^{\infty} \delta(t - nT), \tag{5.12}$$

where $T > 0$ is a constant. Then for any given function $g(t)$, we can calculate the Laplace transform of $g(t)\text{Ш}_T(t)$ as

$$\mathcal{L}[g(t)\text{Ш}_T(t)] = \sum_{n=0}^{\infty} \left(\int_{0^-}^{\infty} \delta(t-nT) g(t) \mathrm{e}^{-st}\, \mathrm{d}t \right) = \sum_{n=0}^{\infty} g(nT)\mathrm{e}^{-snT}. \tag{5.13}$$

[2]This is the Cyrillic letter 'sha' Ш, chosen purely because it looks a bit like a comb.

sequence	$\mathcal{Z}$-transform
a_n	$A(z)$
$\alpha a_n + \beta b_n$	$\alpha A(z) + \beta B(z)$
$\lambda^n a_n$	$A\left(\frac{z}{\lambda}\right)$
na_n	$-zA'(z)$
$a'_n = \begin{cases} a_{n-k} & n \geq k, \\ 0 & n < k. \end{cases}$	$z^{-k}A(z)$
a_{n+1}	$zA(z) - a_0 z$
a_{n+k}	$z^k A(z) - \sum_{j=0}^{k-1} a_j z^{k-j}$
$(a * b)_n$	$A(z)B(z)$

Table 3: Transformation rules for $\mathcal{Z}$-transforms. Here α, β are arbitrary complex constants, $\lambda \neq 0$ and $k \geq 1$.

Integrating against the Dirac comb has the effect of picking out the value of $g(t)\mathrm{e}^{-st}$ at evenly spaced intervals $t = nT$ for $n \in \mathbb{Z}_{\geq 0}$. If we now let $z \doteq \mathrm{e}^{sT}$, this becomes

$$\mathcal{L}[g(t)\text{Ш}_T(t)] = \sum_{n=0}^{\infty} g(nT)z^{-n}, \tag{5.14}$$

matching the $\mathcal{Z}$-transform of the sequence $a_n = g(nT)$. While we won't use this fact, the relation between z and s provides a way of translating stability results from Laplace transforms to $\mathcal{Z}$-transforms, among other things. If you're interested in this, consider how the region of stability for continuous LTI systems $\{s : \mathrm{Re}(s) < 0\}$ changes under the map $s \mapsto \mathrm{e}^{sT}$ and compare with the results later in this chapter.

5.3 Transformation properties

Yes, this is the third section called this. Yes, you know the drill.

As with the other transforms we've seen thus far, the $\mathcal{Z}$-transform is linear, meaning that

$$\mathcal{Z}[\alpha a + \beta b] = \alpha A(z) + \beta B(z), \tag{5.15}$$

for any complex numbers α and β. Instead of scaling a sequence by multiplying by a constant, we can also consider scaling each term of a sequence $a = (a_n)_{n=0}^{\infty}$ exponentially as $b = (\lambda^n a_n)_{n=0}^{\infty}$. Then the $\mathcal{Z}$-transform of b is given by

$$\mathcal{Z}[\lambda^n a_n] = \sum_{n=0}^{\infty} a_n \lambda^n z^{-n} = \sum_{n=0}^{\infty} a_n \left(\frac{z}{\lambda}\right)^{-n} = A\left(\frac{z}{\lambda}\right). \tag{5.16}$$

Similarly, we could scale the sequence linearly by defining $c = (na_n)_{n=0}^{\infty}$. In this case, the $\mathcal{Z}$-transform of c is related to that of a by

$$\mathcal{Z}[na_n] = \sum_{n=0}^{\infty} na_n z^{-n} = z \sum_{n=0}^{\infty} a_n (nz^{-n-1}) = -z \frac{\mathrm{d}}{\mathrm{d}z}\left(\sum_{n=0}^{\infty} a_n z^{-n}\right) = -zA'(z). \tag{5.17}$$

Here we've recognised nz^{-n-1} as the derivative of $-z^{-n}$ and written the series in such a way to use this fact. This provides the analogue of the Laplace transform identity $\mathcal{L}[tf(t)] = -F'(s)$, only here we also have to introduce another factor of z to match up with the definition of the $\mathcal{Z}$-transform.

We can also consider shifting (or translating) the sequence, but since our sequence is indexed by $\mathbb{Z}_{\geq 0}$, we need to distinguish in which direction we're shifting. First consider delaying the sequence by k-many steps, so that the new sequence is

$$b_n = \begin{cases} a_{n-k} & n \geq k, \\ 0 & n < k. \end{cases} \tag{5.18}$$

Here we've been careful to include k-many zeroes, since a_n is not defined for $n < 0$. Since we are essentially moving the whole sequence k-steps to the right, this is often called a ***right translation***. We will often abuse notation and write $(a_{n-k})_{n=0}^{\infty}$ to denote the right translation of $(a_n)_{n=0}^{\infty}$ by k-many steps with the understanding that we are including k-many zeroes at the start of the sequence. The $\mathcal{Z}$-transform of a right translation is given by

$$\mathcal{Z}[b_n] = \sum_{n=k}^{\infty} a_{n-k} z^{-n} = z^{-k} \sum_{n=k}^{\infty} a_{n-k} z^{-(n-k)}. \tag{5.19}$$

By introducing the variable $m = n - k$, we can rewrite the final sum to get

$$\mathcal{Z}[b_n] = z^{-k} \sum_{m=0}^{\infty} a_m z^{-m} = z^{-k} A(z). \tag{5.20}$$

Therefore, a right translation of k-many steps just introduces an extra factor of z^{-k} to the $\mathcal{Z}$-transform. On the other hand, a ***left translation*** $b_n = (a_{n+k})_{n=0}^{\infty}$ is a little more complicated. In particular, the $\mathcal{Z}$-transform is given by

$$\mathcal{Z}[a_{n+k}] = \sum_{n=0}^{\infty} a_{n+k} z^{-n} = z^k \sum_{n=0}^{\infty} a_{n+k} z^{-(n+k)} = z^k \sum_{m=k}^{\infty} a_m z^{-m}. \tag{5.21}$$

Now we see the issue: we are missing terms including a_n for $n < k$, so we need to add them by hand in order to write this in terms of $A(z)$. This is easily done

$$\mathcal{Z}[a_{n+k}] = z^k \left(\sum_{m=0}^{\infty} a_m z^{-m} - \sum_{m=0}^{k-1} a_m z^{-m} \right) = z^k A(z) - \sum_{m=0}^{k-1} a_m z^{-m+k}. \tag{5.22}$$

For concreteness, let's take $k = 2$, then this rule reads as

$$\mathcal{Z}[a_{n+2}] = z^2 A(z) - z^2 a_0 - z a_1. \tag{5.23}$$

This should remind you of the Laplace transform, where derivatives get mapped to multiplication by s^m up to some boundary terms. For the $\mathcal{Z}$-transform, the role of these boundary terms is played by a_n for $n < k$.

Finally, we also have a discrete version of the convolution theorem.

Theorem 5.1. Given two sequences $(a_n)_{n=0}^{\infty}$ and $(b_n)_{n=0}^{\infty}$, the $\mathcal{Z}$-transform of the discrete convolution

$$(a * b)_k = \sum_{n=0}^{k} a_n b_{k-n}, \tag{5.24}$$

is given by

$$\mathcal{Z}[(a * b)_k] = A(z)B(z). \tag{5.25}$$

Proof. Substituting the series expressions for $A(z)$ and $B(z)$ in $A(z)B(z)$ gives

$$A(z)B(z) = \left(\sum_{n=0}^{\infty} a_n z^{-n}\right)\left(\sum_{m=0}^{\infty} b_m z^{-m}\right) = \sum_{n=0}^{\infty}\sum_{m=0}^{\infty} a_n b_m z^{-n-m}. \tag{5.26}$$

By introducing the variable $k = m + n$, we can rewrite the sum as

$$A(z)B(z) = \sum_{k=0}^{\infty}\left(\sum_{n=0}^{k} a_n b_{k-n}\right) z^{-k}. \tag{5.27}$$

We recognise the term in the bracket as $(a * b)_k$, so we identify this with the $\mathcal{Z}$-transform of $(a * b)$. □

Example 5.2. (i) We will use the transformation properties above to derive the $\mathcal{Z}$-transform of the sequence

$$a_n = \begin{cases} 1/n & n \geq 1, \\ 0 & n = 0. \end{cases} \tag{5.28}$$

First notice that the sequence $(na_n) = (0, 1, 1, \ldots)$ is the right-translation of the constant sequence $u = (1, 1, 1, \ldots)$. Therefore,

$$-zA'(z) = \mathcal{Z}[na_n] = z^{-1}\mathcal{Z}[u_n] = \frac{1}{z-1}. \tag{5.29}$$

Rearranging for $A'(z)$, we get

$$A'(z) = -\frac{1}{z(z-1)} = \frac{1}{z} - \frac{1}{z-1,} \tag{5.30}$$

which, upon integration, gives

$$A(z) = C + \log z - \log(z-1) = C + \log\left(\frac{z}{z-1}\right). \tag{5.31}$$

In order to determine the constant of integration C, we need to use the fact that $a_0 = 0$ somehow. We can do so by noticing that $A(z) \to a_0$ as $z \to \infty$. Since $\log(z/(z-1)) \to 0$ as

$z \to \infty$, we have that $C = 0$. All in all, we have

$$\mathcal{Z}[a_n] = \log\left(\frac{z}{z-1}\right) = -\log(1 - z^{-1}). \tag{5.32}$$

(ii) Consider the constant sequence $a = (1)_{n=0}^{\infty}$ and $b = (n)_{n=0}^{\infty}$. The convolution of these sequences is

$$(b * a)_n = \sum_{k=0}^{n} k = \tfrac{1}{2}n(n+1). \tag{5.33}$$

The convolution theorem therefore states that

$$\mathcal{Z}\left[\tfrac{1}{2}(n^2+n)\right] = \frac{z}{z-1} \cdot \frac{z}{(z-1)^2} = \frac{z^2}{(z-1)^3}. \tag{5.34}$$

Now using the linearity of the $\mathcal{Z}$-transform we can rearrange for $\mathcal{Z}[n^2]$ to get

$$\mathcal{Z}[n^2] = \frac{2z^2}{(z-1)^3} - \mathcal{Z}[n] = \frac{2z^2}{(z-1)^3} - \frac{z}{(z-1)^2} = \frac{z^2+z}{(z-1)^3}. \tag{5.35}$$

As an exercise, you can check that this matches the result you'd obtain by using the identity $\mathcal{Z}[n^2] = -zB'(z)$.

5.4 Difference equations

The discrete counterpart to a differential equation is a *difference equation*. These relate the value a_n of a sequence $(a_n)_{n=0}^{\infty}$ to other terms in the sequence. For example, the Fibonacci sequence $(f_n)_{n=0}^{\infty}$ is defined by the difference equation

$$f_{n+2} = f_{n+1} + f_n, \tag{5.36}$$

subject to the conditions $f_0 = 0$ and $f_1 = 1$. More generally, difference equations are useful in modelling systems whose inputs and outputs update in discrete jumps, such as in digital signal processing or population growth models. In the latter, the discrete jumps can represent different generations of a species, with the value of a sequence a_n modelling the population of said species in the nth generation.

So how do we solve difference equations? As you've probably guessed, we can use the $\mathcal{Z}$-transform together with the rules for left and right translations to turn this into a much simpler problem. Taking the Fibonacci example, we have

$$\mathcal{Z}[f_{n+2}] = \mathcal{Z}[f_{n+1}] + \mathcal{Z}[f_n] \implies z^2F(z) - z = zF(z) + F(z). \tag{5.37}$$

Rearranging for $F(z)$ then gives

$$F(z) = \frac{z}{z^2 - z - 1} \tag{5.38}$$

Now we need to invert this to find the general formula for f_n. Just like the continuous case, partial fractions is the key here, but there's a nice trick that we can use to make things a little easier. Recall that we know the $\mathcal{Z}$-transforms

$$\mathcal{Z}[\lambda^n] = \frac{z}{z-\lambda}, \quad \mathcal{Z}[n\lambda^n] = \frac{\lambda z}{(z-\lambda)^2}, \tag{5.39}$$

and so on. The fact that z appears in the numerator of all of these transforms is giving us the hint that we want to write the partial fractions decomposition with a factor of z out front. Therefore, we write the partial fractions decomposition of $F(z)$ as

$$F(z) = z\left(\frac{A}{z-\alpha} + \frac{B}{z-\beta}\right), \tag{5.40}$$

with α and β the distinct roots of $z^2 - z - 1 = 0$. Then comparing this with the $\mathcal{Z}$-transforms above, we have the general solution

$$f_n = A\alpha^n + B\beta^n, \tag{5.41}$$

where A and B are the coefficients obtained from the partial fractions procedure. As an exercise, you can find A, B, α and β and show that

$$f_n = \frac{\varphi^n - \tilde{\varphi}^n}{\sqrt{5}}, \qquad \varphi = \frac{1+\sqrt{5}}{2}, \quad \tilde{\varphi} = \frac{1-\sqrt{5}}{2}. \tag{5.42}$$

This is known as ***Binet's formula*** and it relates the Fibonacci numbers to the golden ratio φ and its conjugate $\tilde{\varphi} = -1/\varphi$.

Example 5.3. Let $(a_n)_{n=0}^{\infty}$ be a sequence that satisfies the following difference equation

$$a_{n+2} + 3a_{n+1} + 2a_n = 0, \tag{5.43}$$

with initial conditions $a_0 = a_1 = 1$. Taking the $\mathcal{Z}$-transform gives

$$z^2A(z) - z^2 - z + 3(zA(z) - z) + 2A(z) = 0, \tag{5.44}$$

which we rearrange to get

$$A(z)(z^2 + 3z + 2) = z^2 + 4z \Longrightarrow A(z) = z\left(\frac{z+4}{z^2+3z+2}\right). \tag{5.45}$$

We can factorise the denominator as $(z+1)(z+2)$ and perform partial fractions on the bracketed part as

$$\frac{z+4}{(z+1)(z+2)} = \frac{A}{z+1} + \frac{B}{z+2} \Longrightarrow z + 4 = A(z+2) + B(z+1). \tag{5.46}$$

Taking $z = -1$ causes the term including B to vanish, while taking $z = -2$ causes the A term to vanish. Therefore,

$$(z=-1): \quad A = 3 \qquad\qquad (z=-2): \quad B = -2. \tag{5.47}$$

As such, the partial fraction decomposition of $A(z)$ is

$$A(z) = \frac{3z}{z+1} - \frac{2z}{z+2}. \tag{5.48}$$

Comparing with the $\mathcal{Z}$-transforms we derived earlier (or on the formula sheet), we obtain

$$a_n = 3(-1)^n - 2(-2)^n. \tag{5.49}$$

We can check the initial conditions as a sanity check

$$a_0 = 3 \cdot 1 - 2 \cdot 1 = 1, \qquad a_1 = 3(-1) - 2(-2) = 1, \tag{5.50}$$

as expected.

Example 5.4. We can use the $\mathcal{Z}$-transform to solve systems of difference equations. Let $a = (a_n)_{n=0}^{\infty}$ and $b = (b_n)_{n=0}^{\infty}$ be two sequences satisfying the difference equations

$$\begin{aligned} a_{n+1} &= 2a_n - b_n, \\ b_{n+1} &= -a_n + b_n, \end{aligned} \tag{5.51}$$

with $a_0 = 0$ and $b_0 = 1$. Taking the $\mathcal{Z}$-transform of both equations gives a set of linear equations for $A(z)$ and $B(z)$

$$\begin{aligned} zA(z) &= 2A(z) - B(z), \\ zB(z) - z &= -A(z) + B(z). \end{aligned} \tag{5.52}$$

Rearranging the top equation for $A(z)$ and substituting this into the bottom equation gives

$$A(z) = -\frac{B(z)}{z-2} \implies B(z)(z-1) = z + \frac{B(z)}{z-2}. \tag{5.53}$$

Now we can rearrange for $B(z)$ to get

$$B(z) = z\left(\frac{z-2}{z^2-3z+1}\right) = z\left(\frac{C}{z-\alpha} + \frac{D}{z-\alpha^{-1}}\right), \tag{5.54}$$

where $\alpha = \frac{1}{2}(3+\sqrt{5})$ is the larger root of $z^2 - 3z + 1 = 0$. As a somewhat annoying exercise, you can show that

$$C = \frac{5-\sqrt{5}}{10}, \qquad D = \frac{5+\sqrt{5}}{10}, \tag{5.55}$$

giving

$$B(z) = \frac{5-\sqrt{5}}{10}\frac{z}{z-\alpha} + \frac{5+\sqrt{5}}{10}\frac{z}{z-\alpha^{-1}}. \tag{5.56}$$

n	0	1	2	3	4	5	6	7	8
a_n	0	−1	−3	−8	−21	−55	−144	−377	−987
b_n	1	1	2	5	13	34	89	233	610

Table 4: The solutions to the system of difference equations in eq. (5.51) for $0 \leq n \leq 8$.

Therefore, the sequence b is given by

$$b_n = \frac{5-\sqrt{5}}{10}\alpha^n + \frac{5+\sqrt{5}}{10}\alpha^{-n}. \tag{5.57}$$

We're not done though, we need to find a_n as well. We know that $A(z) = -B(z)/(z-2)$, so we have

$$A(z) = -z\left(\frac{1}{z^2-3z+1}\right) = \frac{1}{\sqrt{5}}\frac{z}{z-\alpha^{-1}} - \frac{1}{\sqrt{5}}\frac{z}{z-\alpha}. \tag{5.58}$$

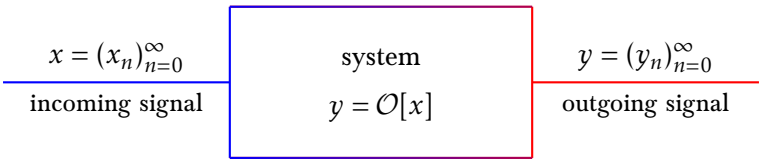
Therefore, the sequence a is given by

$$a_n = \frac{1}{\sqrt{5}}\alpha^{-n} - \frac{1}{\sqrt{5}}\alpha^n. \tag{5.59}$$

This all looks a bit complicated, but calculating the first few values of a_n and b_n displays a nice pattern, as shown in table 4. (Compare these with the first 16 Fibonacci numbers!)

5.5 Return of LTI systems

We now consider the discrete version of LTI systems, which we visualise as a black box as before, with an incoming signal x_n being mapped to the outgoing signal y_n through an operation $\mathcal{O} : x_n \mapsto y_n$.



The requirement that a system is LTI and causal is completely analogous to the continuous case:

- *Linearity.* For any two incoming signals x_n and x'_n with corresponding outgoing signals y_n and y'_n and any choice of constants $\alpha, \beta \in \mathbb{C}$, $\mathcal{O}$ satisfies

$$\mathcal{O}[\alpha x_n + \beta x'_n] = \alpha\mathcal{O}[x_n] + \beta\mathcal{O}[x'_n] = \alpha y_n + \beta y'_n. \tag{5.60}$$

- *Time invariance.* Delaying an incoming signal by k-many steps delays the outgoing signal by k-many steps with no other changes. That is,

$$\mathcal{O}[x_{n-k}] = y_{n-k}, \tag{5.61}$$

  for any incoming signal x_n.

- *Causality.* If x and x' are two incoming signals such that $x_k = x'_k$ for $0 \leq k \leq n$, then the corresponding outgoing signals satisfy $y_n = y'_n$.

The last point is slightly different to how we phrased causality in the continuous case, but it is equivalent. Essentially, since the two incoming signals match up to a certain point, the system cannot distinguish them as it only knows about past values of the incoming signal and knows nothing about how they differ in the future.

Theorem 5.2. The output of any causal discrete LTI system is given by the convolution

$$(x * g)_n = \sum_{k=0}^{n} x_k g_{n-k}, \tag{5.62}$$

where x_n is the incoming signal and g_n is the impulse response $\mathcal{O}[\delta_n]$.

Proof. Let $x = (x_0, x_1, x_2, \ldots)$ be the incoming signal of a causal LTI system and y the corresponding output. We define a new signal $x' = (x_0, x_1, \ldots, x_n, 0, 0, \ldots)$ which matches x up to the nth step. We can decompose x' into a finite sum of impulses

$$x' = x_0(\delta_m)_{m=0}^{\infty} + x_1(\delta_{m-1})_{m=0}^{\infty} + \ldots + x_n(\delta_{m-n})_{m=0}^{\infty} = \sum_{k=0}^{n} x_k(\delta_{m-k})_{m=0}^{\infty}. \tag{5.63}$$

The nth step of the corresponding output is therefore

$$y'_n = \sum_{k=0}^{n} x_k \mathcal{O}[(\delta_{n-k})_{n=0}^{\infty}] = \sum_{k=0}^{n} x_k g_{n-k} = (x * g)_n, \tag{5.64}$$

where we've used the linearity property in the first equality and the time invariance property to establish the last equality. By causality, this must equal y_n. Since we can do this procedure for any value of n, this gives y_n for every $n \in \mathbb{Z}_{\geq 0}$. □

We now restrict to the class of LTI systems whose outgoing signals are given by the solution to a difference equation

$$a_N y_{n+N} + a_{N-1} y_{n+N-1} + \ldots + a_1 y_{n+1} + a_0 y_n = x_n, \tag{5.65}$$

subject to the conditions

$$y_n = 0, \quad \forall n < N. \tag{5.66}$$

These boundary conditions are required by linearity and state that the outgoing signal is completely at rest until N-many time steps have passed. The nice thing about this assumption is that the $\mathcal{Z}$-transform of eq. (5.65) is particularly simple

$$Y(z)(a_N z^N + a_{N-1} z^{N-1} + \ldots + a_1 z + a_0) = X(z), \tag{5.67}$$

leading to the solution

$$Y(z) = \frac{X(z)}{a_N z^N + a_{N-1} z^{N-1} + \ldots + a_1 z + a_0}. \tag{5.68}$$

When $x = (\delta_n)_{n=0}^{\infty}$, $X(z) = 1$ and we identify the *transfer function* $G(z)$ with the rational function

$$G(z) = \frac{1}{a_N z^N + a_{N-1} z^{N-1} + \ldots + a_1 z + a_0}, \tag{5.69}$$

so $Y(z) = G(z)X(z)$, as expected.

Example 5.5. Consider the LTI system governed by the difference equation

$$y_{n+2} - 4y_{n+1} + 4y_n = x_n, \tag{5.70}$$

subject to $y_0 = y_1 = 0$. Taking the $\mathcal{Z}$-transform, we get

$$Y(z)(z^2 - 4z + 4) = X(z) \Longrightarrow Y(z) = \frac{X(z)}{(z-2)^2}. \tag{5.71}$$

Taking $X(z) = 1$ identifies the transfer function with $1/(z-2)^2$. This almost looks like the $\mathcal{Z}$-transform of the sequence $b = (n2^{n-1})_{n=0}^{\infty}$, but we're missing the factor of z in the numerator. By looking at the formula sheet, we see that this corresponds to an additional right translation by 1 step, so the inverse sequence is

$$g_n = \begin{cases} (n-1)2^{n-2} & n \geq 1, \\ 0, & n = 0. \end{cases} \tag{5.72}$$

The first few terms of the sequence are therefore $g = (0, 0, 1, 4, 12, \ldots)$. Therefore, the full solution for y_n is given by

$$y_n = \sum_{k=0}^{n} x_k g_{n-k} = \sum_{k=0}^{n-1} x_k (n-k-1) 2^{n-k-2}. \tag{5.73}$$

Here the upper limit is $n-1$, since $g_0 = 0$. For example, if $x_k = 2^k$, the corresponding solution is

$$y_n = \sum_{k=0}^{n-1} (n-k-1) 2^{n-2} = 2^{n-2} \sum_{k=0}^{n-1} (n-k-1). \tag{5.74}$$

It can be shown that the final sum equates to $n(n-1)/2$, giving the solution $y_n = 2^{n-3}n(n-1)$. As a check, we can substitute this into the difference equation with $n > 0$ to get

$$\begin{aligned} y_{n+2} - 4y_{n+1} + 4y_n &= 2^{n-1}(n+2)(n+1) - 2^n(n+1)n + 2^{n-1}n(n-1), \\ &= 2^{n-1}(n^2 + 3n + 2 - 2(n^2+n) + n^2 - n), \\ &= 2^{n-1} \cdot 2, \\ &= 2^n, \end{aligned} \tag{5.75}$$

verifying the solution.

5.6 Stability analysis

Now we turn to the stability of discrete LTI systems, the definition of which is much the same as before, just adapted to sequences.

Definition 5.2. (i) A discrete LTI system is *stable* if

$$\lim_{n\to\infty} g_n = 0, \tag{5.76}$$

where g is the impulse response.

(ii) The system is ***bounded-input, bounded-output (BIBO)*** stable if any bounded incoming signal results in a bounded outgoing signal. That is, for any incoming x_n satisfying $|x_n| \leq A$ for all n, there exists $B \geq 0$ such that $|y_n| \leq B$ for all n.

One can reformulate BIBO stability as

$$\|g\|_1 = \sum_{n=0}^{\infty} |g_n| < \infty. \tag{5.77}$$

This is equivalent to our definition since, assuming that $|x_n| \leq A$ for some constant A, we have that the outgoing signal satisfies

$$|y_n| \leq \sum_{k=0}^{n} |x_k||g_{n-k}| \leq A \sum_{k=0}^{n} |g_{n-k}| = A \sum_{m=0}^{n} |g_m| \leq A\|g\|_1, \tag{5.78}$$

where we've defined $m = n-k$ in the middle equality. Demanding that $\|g\|_1$ is finite then implies that y_n is bounded, showing that $\|g\|_1 < \infty$ implies BIBO stability as defined above. Similarly, it is easy to show that assuming bounded inputs and outputs demands $\|g\|_1 < \infty$, showing the equivalence of the two definitions.

Again, for LTI systems with rational transfer functions these are equivalent notions, but for non-rational transfer functions they are not. Restricting now to rational transfer functions, we have the following result.

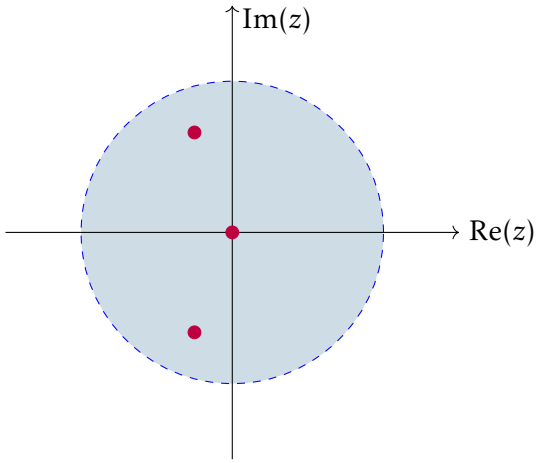


Figure 9: The poles of the transfer function $G(z)$ in eq. (5.80).

Theorem 5.3. A discrete causal LTI system with rational transfer function $G(z)$ is stable if and only if the poles of $G(z)$ belong to the set $\mathbb{D} = \{z : |z| < 1\}$.

Example 5.6. Consider the LTI system governed by the difference equation

$$2y_{n+3} + y_{n+2} + y_{n+1} = x_n, \tag{5.79}$$

subject to $y_0 = y_1 = y_2 = 0$. Transforming this equation leads to

$$Y(z)(2z^3 + z^2 + z) = X(z) \Longrightarrow G(z) = \frac{1}{z(2z^2 + z + 1)}. \tag{5.80}$$

Here we've identified the transfer function $G(z)$ by setting $X(z) = 1$. The poles of $G(z)$ occur when $z = 0$ and $2z^2 + z + 1 = 0$. Using the quadratic formula on the latter gives

$$z_\pm = \frac{-1 \pm \sqrt{1 - 4 \cdot 2 \cdot 1}}{4} = -\frac{1}{4} \pm \mathrm{i}\frac{\sqrt{7}}{4}. \tag{5.81}$$

In order to check the stability of the system, we need to see if the poles have modulus less than 1. The pole at $z = 0$ is clearly fine, while the modulus of $z_\pm$ is given by

$$|z_\pm| = \sqrt{\frac{1}{16} + \frac{7}{16}} = \frac{1}{\sqrt{2}} < 1. \tag{5.82}$$

Therefore, the system is stable.